

# 2

CHAPTER

# Acids, Bases and Salts



## School Level

### Introduction

The two main sources of most of the chemical substances are:

- (1) Animals and plants                      (2) Minerals and rocks

The chemical substances obtained from animals and plants (living organisms) are called **organic compounds** whereas those obtained from minerals and rocks are called **inorganic compounds**. These compounds are present in the food stuff as well as in a number of other materials that we use in everyday life like washing soda, lime etc.

The earliest classification of inorganic compounds, including a large number of organic compounds, was on the basis of their taste. On this basis, the compounds were classified into the following three categories:

- (i) Acids            (ii) Bases and            (iii) Salts

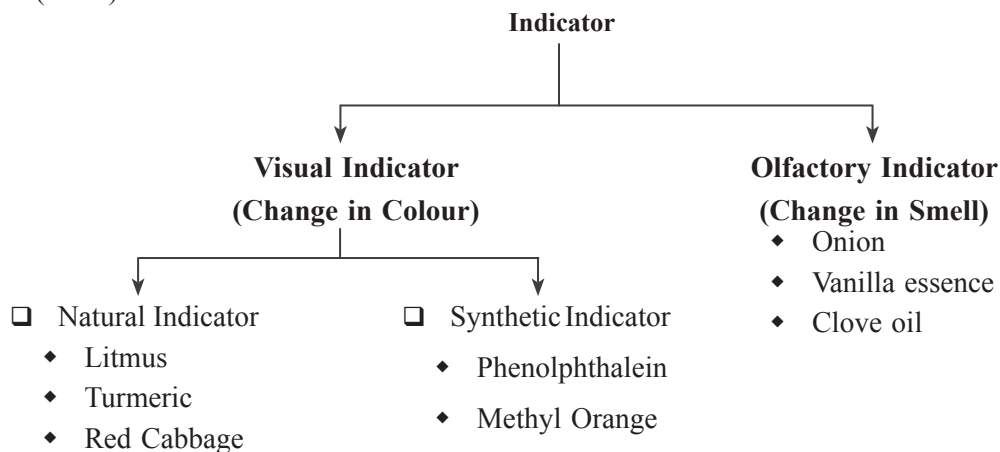
**Acids** are the substances which have sour taste. The term has been derived from the Latin word, *acidus*, which means sour. A few examples of acidic substances used in everyday life include lemon juice, tomatoes, vinegar etc.

**Bases** are the substances which have bitter taste. A few examples of substances having basic nature include washing soda, baking soda etc.

**Salts** are the compounds formed from acids and bases. A few examples of the salts include sodium chloride, copper sulphate, silver nitrate, calcium carbonate etc. Salts having taste similar to that of common salt are called salty. However, some salts have very unpleasant taste and even may be poisonous. Hence, it is very dangerous to taste any chemical substance. You are therefore, advised never to taste any chemical substance.

### Indicators

An indicator is a chemical compound which indicates the presence of acidic, basic or neutral substance either by change in colour or odour(smell).



A few common acid-base indicator are given below:

- ❑ **Litmus:** It is a natural dye made from lichens (a plant belong to division Thallophyta). Litmus solution is a purple dye that is used as an indicator. Blue and red litmus solutions are prepared from two different varieties of lichens.
- ❑ **Litmus Paper:** Blue or red litmus paper is made by dipping a strip of filter paper in blue or red litmus solutions. The paper is then removed from the solution and then dried up. Blue litmus paper will turn red in an acidic solution and red litmus paper will turn blue in a basic solution.
- ❑ **Turmeric Juice:** It has yellow colour. It is yellow in acidic or neutral solutions but changes to deep brown colour in a basic solution.
- ❑ **Red-cabbage Juice:** It has purple colour, it changes to red colour in an acidic solution, but green in a basic solution.

## Synthetic Indicators

- ❑ **Methyl Orange:** Solid methyl orange in very little amount is dissolved in hot water and then filtered. The filtrate is used as methyl orange that will turn red in acidic solutions and yellow in basic solutions.
- ❑ **Phenolphthalein:** An alcoholic solution of phenolphthalein is used as an indicator. It become colourless in an acidic solution, but it will become pink in basic solution.

**Table-1:** Some acid-base indicators in acid, base and neutral solution are given below:

| Indicator         | Acid solution | Basic solution | Neutral solution |
|-------------------|---------------|----------------|------------------|
| Red litmus        | No change     | Blue           | No change        |
| Blue litmus       | Red           | No change      | No change        |
| Turmeric          | Yellow        | Reddish brown  | Yellow           |
| Red cabbage juice | Red           | Green          | Purple           |
| Phenolphthalein   | Colourless    | Pink           | Colourless       |
| Methyl Orange     | Red           | Yellow         | Orange           |

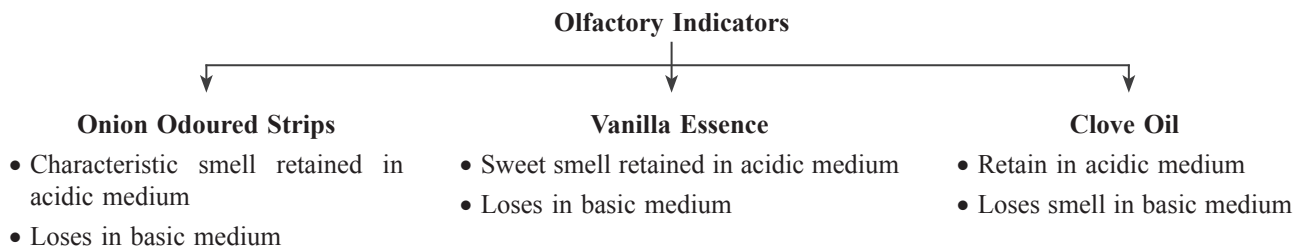
**Table-2:** The indicators are used to test whether some of the substances of daily use as given below that are acidic or basic.

| Acidic substances | Basic substance        |
|-------------------|------------------------|
| Vitamin C tablets | Soap solution          |
| Tomato juice      | Toothpaste             |
| Vinegar           | Antacids               |
| Lemon juice       | Washing soda solutions |
| Orange juice      | Bleach                 |

## Olfactory Indicators

Some substances whose odour change in acidic or basic solutions are called olfactory indicators.

**For example:** Vanilla extract, onion and clove oil, etc.



## Activity

**Aim:** To test acids and bases in the laboratory using acid-base indicators.

**Procedure:** Collect the following samples from the science laboratory-hydrochloric acid (HCl), sulphuric acid (H<sub>2</sub>SO<sub>4</sub>), Nitric acid (HNO<sub>3</sub>), acetic acid (CH<sub>3</sub>COOH), sodium hydroxide (NaOH), calcium hydroxide [Ca(OH)<sub>2</sub>], potassium hydroxide (KOH), magnesium hydroxide [Mg(OH)<sub>2</sub>] and ammonium hydroxide (NH<sub>4</sub>OH).

- ❑ Put a drop of each of the above solutions on a watch-glass and test with a drop of the following indicators. Red litmus solution. Blue litmus solution. Phenolphthalein solution. Methyl orange solution.

**Observation:** The changes observed are as follow:

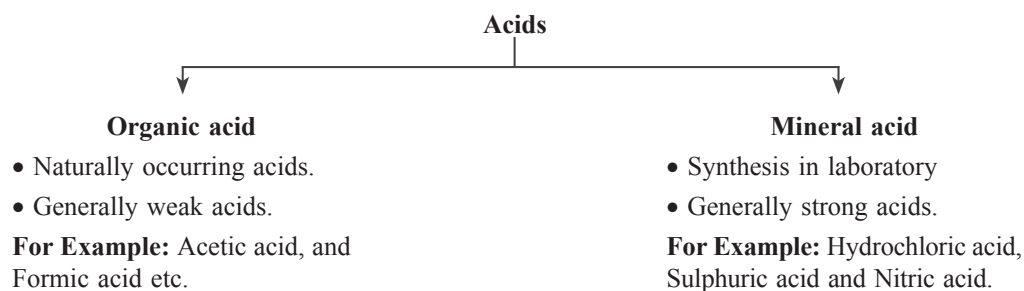
|    | Sample Solution                                  | Red litmus solution | Blue litmus solution | Phenolphthalein solution | Methyl orange solution |
|----|--------------------------------------------------|---------------------|----------------------|--------------------------|------------------------|
| 1. | Hydrochloric acid (HCl).                         | No effect           | Red                  | Colourless               | Red                    |
| 2. | Sulphuric acid (H <sub>2</sub> SO <sub>4</sub> ) | No effect           | Red                  | Colourless               | Red                    |
| 3. | Nitric acid (HNO <sub>3</sub> )                  | No effect           | Red                  | Colourless               | Red                    |
| 4. | Acetic acid (CH <sub>3</sub> COOH)               | No effect           | Red                  | Colourless               | Red                    |
| 5. | Sodium hydroxide (NaOH)                          | Blue                | No effect            | Pink                     | Yellow                 |
| 6. | Calcium hydroxide, (Ca(OH) <sub>2</sub> )        | Blue                | No effect            | Pink                     | Yellow                 |
| 7. | Potassium hydroxide (KOH)                        | Blue                | No effect            | Pink                     | Yellow                 |
| 8. | Magnesium hydroxide, Mg(OH) <sub>2</sub>         | Blue                | No effect            | Pink                     | Yellow                 |
| 9. | Ammonium hydroxide (NH <sub>4</sub> OH)          | Blue                | No effect            | Pink                     | Yellow                 |

**Conclusion:** Acids turn blue litmus red but have no effect on red litmus. Bases turn red litmus blue but have no effect on blue litmus. Phenolphthalein is colourless in acidic medium and turns pink in basic medium. Methyl orange is yellow in basic medium and red in acidic medium.

## Acids

An acid is defined as a substance that release one or more Hydronium ion or H<sup>+</sup> ions in an aqueous solution.

**For example:** When HCl is dissolved in water, it gives H<sup>+</sup> and Cl<sup>-</sup>, this H<sup>+</sup> ion cannot exist individually and combine with a H<sub>2</sub>O molecule and form a H<sub>3</sub>O<sup>+</sup> ion.



## Knowledge Hub

- ❖ Hydrogen is common in all acids but not all the molecules that contain hydrogen are necessarily an acid. For example CH<sub>4</sub>, NH<sub>3</sub> etc., have hydrogen atoms but these are not acids.

## Physical Properties of the Acids

- ❑ Acids are generally sour taste.
- ❑ Acids turns blue litmus red and methyl orange to red.
- ❑ Acids are highly corrosive in nature.
- ❑ Acids in their aqueous solutions conduct electricity.

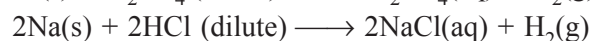
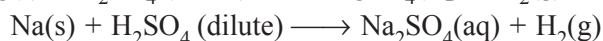
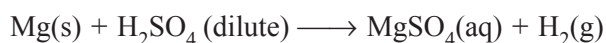
## Chemical Properties of the Acids

### Reaction of Acids with Metals

- ❑ Dilute acids like dilute HCl and dilute  $\text{H}_2\text{SO}_4$  both react with active metals to release hydrogen gas.



For Example



### Knowledge Hub

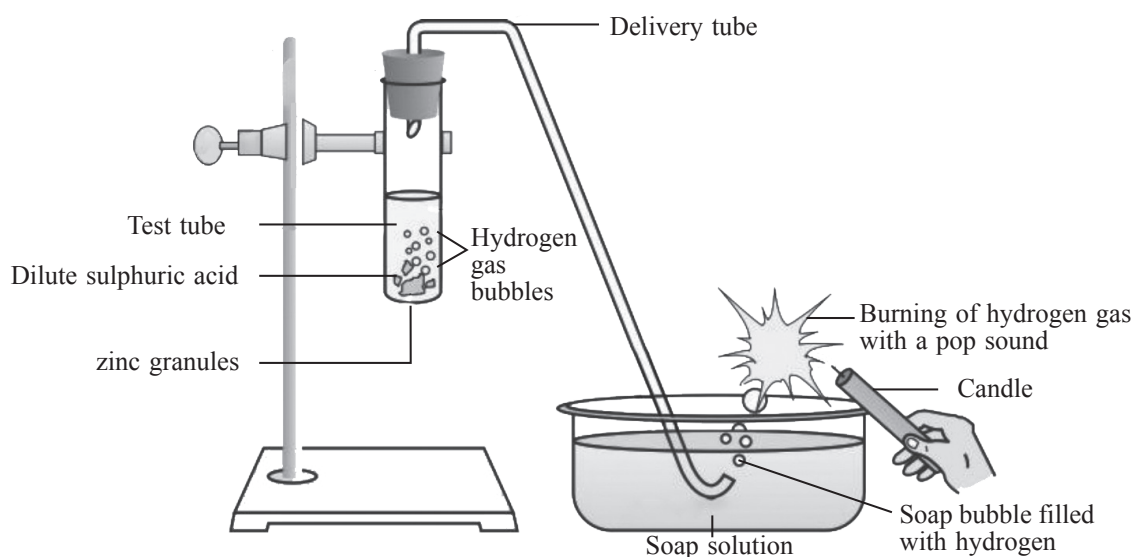
- ❖ Hydrogen gas is not evolved when a metal reacts with nitric acid because  $\text{HNO}_3$  is a strong oxidising agent and it oxidises the  $\text{H}_2$  produced to water and itself gets reduced to any of the nitrogen oxides ( $\text{N}_2\text{O}$ ,  $\text{NO}$ ,  $\text{NO}_2$ ). But Mg and Mn reacts with very dilute  $\text{HNO}_3$  to evolve  $\text{H}_2$  gas.

## Activity

**Aim:** To study the reaction of acids with metals.

**Procedure:**

- ❑ Set the apparatus as shown in Fig below.
- ❑ Take about 5mL of dilute sulphuric acid in a test tube and add a few pieces of zinc granules to it.
- ❑ Pass the gas being evolved through the soap solution.



**Note:** Granulated zinc is used as it increases the rate of reaction.

**Observation:** Bubbles are formed in the soap solution indicating that some gas is evolved during the reaction between the metal and the acid. The gas burns with a popping sound showing that the gas evolved is hydrogen.

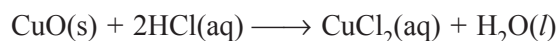
**Conclusion:** The active metal like zinc displaces hydrogen from the acid to form the corresponding salt and hydrogen gas.



## Reaction of Acids with Metal Oxides

- ❑ Acids react with metal oxides to produce salt and water.
- ❑ Metal oxides are basic in nature.

**For Example**

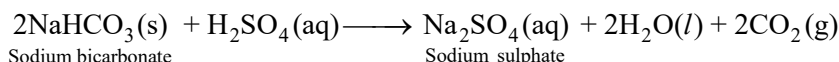
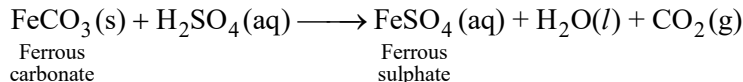
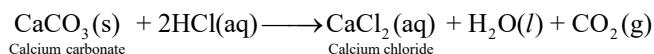


## Reaction of Acid with Metal Carbonates and Metal Bicarbonates

Both metal carbonates and bicarbonates react with acids to release  $\text{CO}_2$  gas and form salts and water.



**For Example:**

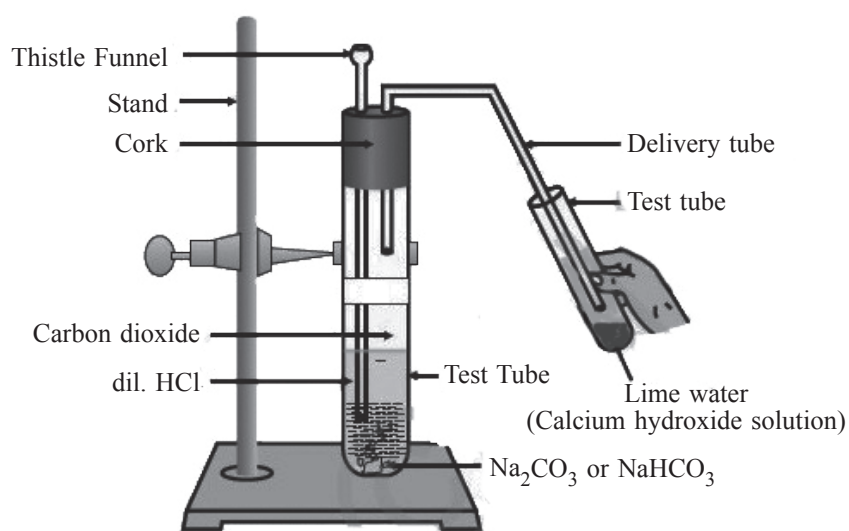


## Activity

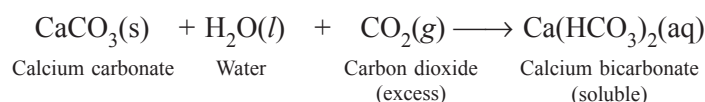
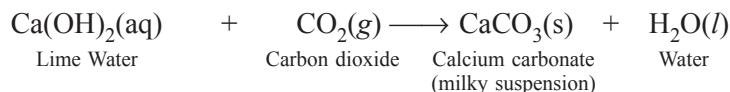
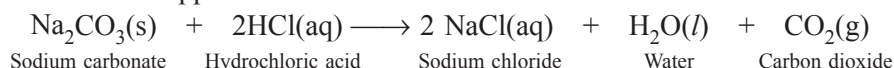
**Aim:** To study the reaction of acids with metal carbonates and metal bicarbonates.

**Procedure:**

- ❑ Set the apparatus as shown in Fig below.
- ❑ Take sodium carbonate in test tube.
- ❑ Add dilute hydrochloric acid through the thistle funnel. Allow the gas evolved to pass through lime water taken in a test tube.



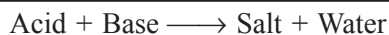
**Observation:** The reaction of sodium carbonate with dilute hydrochloric acid by passing the gas evolved through lime water turns milky, showing that the gas evolved is carbon dioxide. If excess of  $\text{CO}_2$  gas is passed through lime water, milkiness disappears and the solution becomes clear due to the formation of soluble calcium bicarbonate.



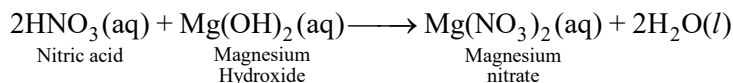
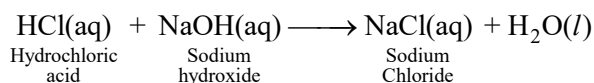
**Note:** Calcium carbonate is main constituent of chalk / marble / limestone/ egg shell.

## Reaction of Acids with Bases

The reaction between an acid and a base to give a salt and water is known as neutralisation reaction.



**For Example**



## Activity

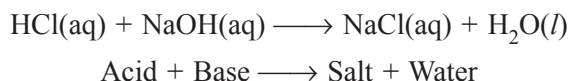
**Aim:** To study the reaction between acids and bases, *i.e.*, neutralisation reaction.

**Procedure:**

- ☐ Take about 2mL of dilute NaOH solution in a test tube and add two drops of phenolphthalein solution.
- ☐ Note the colour of the solution.
- ☐ Add dilute HCl solution to the above solution drop by drop.
- ☐ Note the change in colour of the reaction mixture.
- ☐ Now add a few drops of NaOH solution to the above mixture.

**Observation:** On adding phenolphthalein to NaOH solution, the colour becomes pink. On adding dilute HCl solution drop wise, finally the pink colour disappears and the solution again becomes colourless. On adding, NaOH pink colour reappears because the medium becomes basic again.

**Conclusion:** We have concluded that the effect of a base is nullified by an acid and vice-versa. The reaction taking place is written as



## Uses of Acids

| Acids                                      | Uses                                                                                                                                                                                                                                                      |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Boric acid ( $\text{H}_3\text{BO}_3$ )     | <input type="checkbox"/> In the manufacturing of glass and enamels, leather, paper and explosives.<br><input type="checkbox"/> Used as preservative for grains.<br><input type="checkbox"/> Used in detergents.                                           |
| Nitric acid ( $\text{HNO}_3$ )             | <input type="checkbox"/> Nitric acid that is present in rainwater forms nitrates in the soil that are then used by plants to obtain nitrogen.<br><input type="checkbox"/> In the manufacturing of fertilizers like ammonium nitrate, explosives like TNT. |
| Sulphuric acid ( $\text{H}_2\text{SO}_4$ ) | <input type="checkbox"/> In the manufacture of detergents and artificial fibre.<br><input type="checkbox"/> In storage batteries.<br><input type="checkbox"/> In the manufacturing of hydrochloric acid and alum.                                         |
| Hydrochloric acid ( $\text{HCl}$ )         | <input type="checkbox"/> It is present in the gastric juice which helps in the digestion of food.<br><input type="checkbox"/> As a bathroom cleaner.<br><input type="checkbox"/> Used in polyvinyl chloride (PVC) manufacture.                            |

## Base

- ☐ A base is defined as a substance that are capable of releasing one or more  $\text{OH}^-$  ions in an aqueous solution.
- ☐ Substances that are bitter in taste and soapy in touch are called as bases.

**For Example:** Sodium hydroxide, potassium hydroxide, Aluminum hydroxide and Ammonium hydroxide.

## Alkalies

Alkalies are defined as the water soluble bases. Some bases like sodium hydroxide and potassium hydroxide are water soluble. These are called as alkalies.

- ☐ Bases like  $\text{Cu}(\text{OH})_2$ ,  $\text{Fe}(\text{OH})_3$  and  $\text{Al}(\text{OH})_3$ , these are not alkalies.

**Table-3:** A few typical bases along with their chemical formulae and uses are given below:

| Name of base        | Chemical Formula         | Commercial Name  | Uses                                                                                   |
|---------------------|--------------------------|------------------|----------------------------------------------------------------------------------------|
| Potassium hydroxide | $\text{KOH}$             | Caustic potash   | Storage batteries, in manufacturing of soap, absorbing $\text{CO}_2$ gas, etc.         |
| Calcium hydroxide   | $\text{Ca}(\text{OH})_2$ | Slaked lime      | In manufacturing of bleaching powder, used in softening of hard water etc.             |
| Aluminum hydroxide  | $\text{Al}(\text{OH})_3$ | —                | As foaming agent in fire extinguishers.                                                |
| Sodium hydroxide    | $\text{NaOH}$            | Caustic Soda     | In manufacture of soap, paper, pulp, refining process of petroleum etc.                |
| Magnesium hydroxide | $\text{Mg}(\text{OH})_2$ | Milk of Magnesia | Used as laxative. As an antacids to remove acidity from stomach.                       |
| Ammonium hydroxide  | $\text{NH}_4\text{OH}$   | —                | In cleaning window panes, In removing greases stains from clothes, used as fertilizer. |

## Physical Properties of Bases

- ❑ Bases are generally bitter in taste and soapy in touch.
- ❑ Bases turn red litmus blue and methyl orange to yellow.
- ❑ Bases are highly corrosive in nature.
- ❑ Bases in their aqueous solutions conduct electricity.

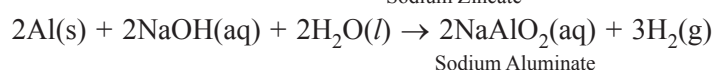
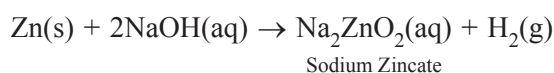
## Chemical Properties of Bases

### Reaction of Bases with Metals.

- ❑ Metal like zinc and aluminium react with strong alkalies like NaOH (Caustic Soda), KOH (Caustic Potash) to evolve hydrogen gas.



For Example

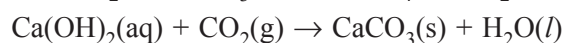
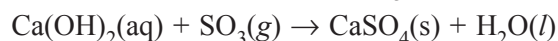
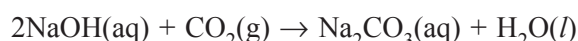


### Reaction of Bases with Non-metallic Oxides

- ❑ Bases react with Non-metallic Oxides to produce Salt and water.



For Example



## Dilution of Acids and Bases

- ❑ Acids and bases are mostly soluble in water and they can be diluted by adding the required amount of water.
- ❑ The amount of acid or base per unit volume decreases with the addition of water. This process is generally exothermic in nature.

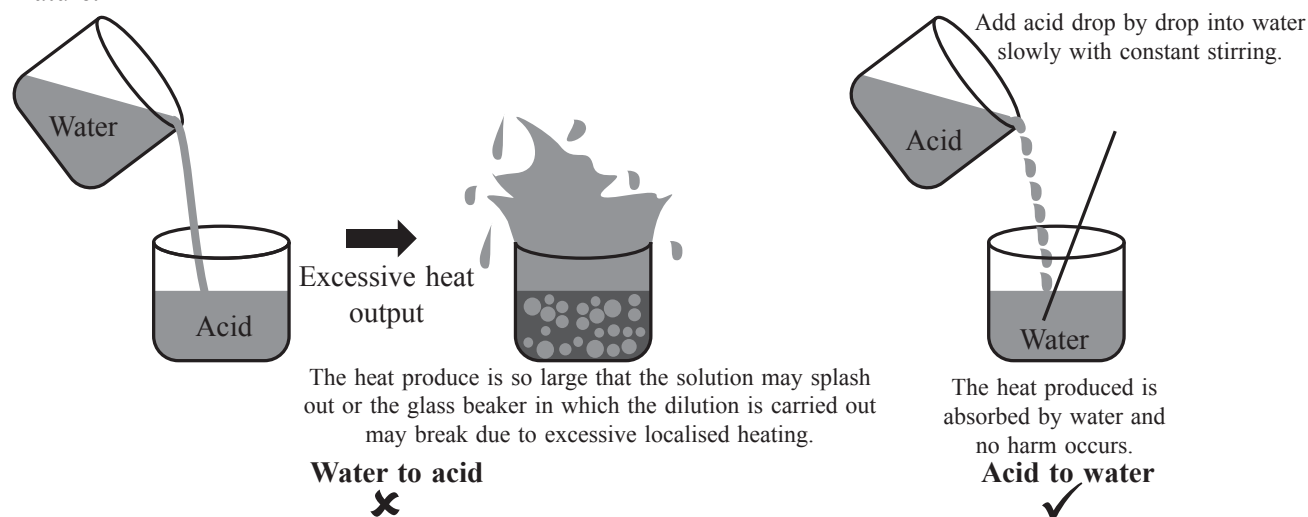


Fig. 1: Dilution of acids and bases.



## Try it Yourself

1. What will happen when egg shell is placed in Conc.  $\text{HNO}_3$ ?
2. Why should curd and sour substances not be kept in brass vessel?
3. Do basic solutions also have  $\text{H}^+(\text{aq})$  ions? If yes, then why are these basic?

## Strength of Acids and Bases

### pH Scale

A scale that is used to measure the strength of acids and bases in a solution is known pH scale. It has been developed by S.P.L. Sorensen. The P in pH stands for 'potenz' in German meaning power and 'H' stands for hydrogen ions.

- pH should be thought of simply as a number that indicates the acidic or basic strength of solution.
- On the pH scale, we can measure pH generally from 0 (very acidic) to 14 (very alkaline).
- Higher the  $\text{H}^+$  ion concentration, Lower will be value of the pH scale.

The pH of an aqueous solution is the negative logarithm of its  $\text{H}^+$  ion concentration. That is,

$$\text{pH} = -\log [\text{H}^+].$$

Just as the  $[\text{H}^+]$  of a solution can be expressed in terms of pH value, the  $[\text{OH}^-]$  can be expressed as pOH.

$$\text{pOH} = -\log [\text{OH}^-]$$

$$\boxed{\text{pH} + \text{pOH} = 14} \text{ at } 25^\circ\text{C}.$$

### Characteristic of pH scale are

For acidic solution,  $\text{pH} < 7$

For alkaline solution,  $\text{pH} > 7$

For neutral solution,  $\text{pH} = 7$

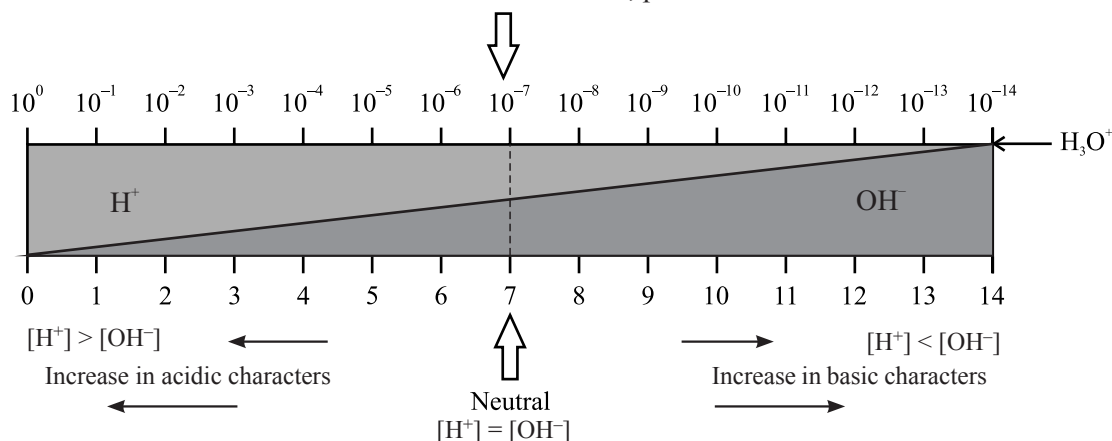


Fig. 2: pH Scale for measuring acidic and basic strength

## Universal Indicator

- A universal indicator is mixture of a number of indicators. It shows different colours at different pH values.

For low pH → Red colour

For neutral → Green colour

For high pH → Blue or violet

- The exact pH of the solution will be measured with the use of pH meter that will give instant reading and it can be relied upon.

## Importance of pH in Everyday Life

pH plays a very important role in our everyday life.

1. **pH of our digestive system:** Hydrochloric acid that is produced in our stomach helps in digestion of food without causing any harm to the stomach. But if the amount of the acid is above a certain limit due to indigestion or eating spicy foods then pain and irritation occur in the stomach.

**Remedy:** In order to neutralize the effect of excess acid, a mild base called antacid is usually taken. Eg: Magnesium hydroxide (milk of magnesia) is a mild base that is usually used as an antacid.

2. **pH change leads to tooth decay:** When we eat high sugar content food, it break down by bacteria present in the mouth and an acid is formed.

When the pH of the mouth becomes lower than 5.5 then tooth enamel (made of calcium phosphate) gets corroded. Saliva, that is slightly alkaline in nature produced in the mouth neutralizes some acid, but excess acid remains unaffected.

**Remedy:** The excess acid is removed only by the use of toothpaste that is alkaline in nature. Neem stick also contains alkaline juice. So, the cleaning of tooth by Neem stick or toothpaste helps in reducing tooth decay.

3. **Self-defence by animals and plants through chemical warfare:** Honey-bee inserts an acid through its stings that causes pain and irritation. This, a mild base like baking soda is applied to treat the wound. Similarly, nettle leaves, that have stinging hairs, when touched insert formic acid in our body. This causes a burning sensation.

The effect of stings by wasps containing alkali is neutralised by the application of vinegar which is dilute solution of ethanoic acid (or acetic acid).

4. **pH change in soil:** Plants require a specific pH range for their healthy growth. Soil with high iron minerals or vegetation tend to become acidic. The soil pH can reach as low as 4. The acidic effect can be neutralised by 'liming the soil'. Which is carried by adding calcium oxide, calcium hydroxide or calcium carbonate. These are basic in nature. Similarly, the soil with excess of lime stone or chalk can reach pH as high as 8.3 and is quite harmful for the plant growth. In order to reduce the alkaline effect, it is better to add some decaying organic matter (compost or manure).



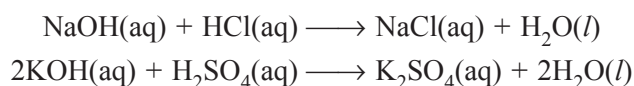
### Try it Yourself

1. What is the pH value of milk when it changes into curd?
2. A student detected the pH of four unknown solution P, Q, R and S as follows 11, 5, 7 and 2. Predict the nature of the solution.
3. You are given three solutions A, B and C with pH values 2, 10 and 13 respectively. Write which solution has more hydrogen ion concentration among the three and state the nature 'acidic or basic' of each solution.

## Salts

A substance formed by neutralisation of an acid with a base is called a salt.

**For Example:**



## Chemicals from Common Salt

### Sodium Chloride (NaCl)-Common Salt

- ❑ Sodium chloride (NaCl) also known as common salt or table salt. It is the most important part of our diet.
- ❑ It is produced by the reaction between of sodium hydroxide (NaOH) and hydrochloric acid (HCl) solutions.

- ❑ Deposits of the salts are found in different part of the world and is called as Rock salt. When salt is pure, it is white in colour & crystalline solid. However, it is often brown in colour due to the presence of impurities in it.

### Properties of Common Salt

- ❑ Crystalline colourless substance
- ❑ Its solubility at room temperature is about 35.8g per 100g of water.
- ❑ It is slightly hygroscopic in nature. This is due to the presence of magnesium and calcium chloride in it.

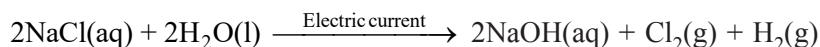
### Uses of NaCl

- 1. Raw material for chemicals:** Sodium chloride is a very useful raw material for different chemicals. Like for hydrochloric acid (HCl), washing soda ( $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ ), baking soda ( $\text{NaHCO}_3$ ) etc.
- 2. Essential for life:** Sodium chloride is quite important for life. Biologically, NaCl has a number of functions to perform such as in muscle contraction, in conduction of nerve impulse in the nervous system and it is also converted in HCl that helps in the digestion of food in the stomach.
- 3. Preservative:** It is used as a preservative for a number of food materials, *e.g.*, in packed meat and fish.
- 4.** It can also be used as a freezing mixture when mixed with ice.

## Sodium Hydroxide (NaOH)

NaOH is chemically known as **Caustic Soda**. It is a strong base.

When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. The process is known as the **chlor-alkali process** because of the products formed-chlor for chlorine and alkali for sodium hydroxide.



- ❑ Chlorine gas is released at the anode.
- ❑ Hydrogen gas is released at the cathode.
- ❑ Sodium hydroxide solution is formed near the cathode.

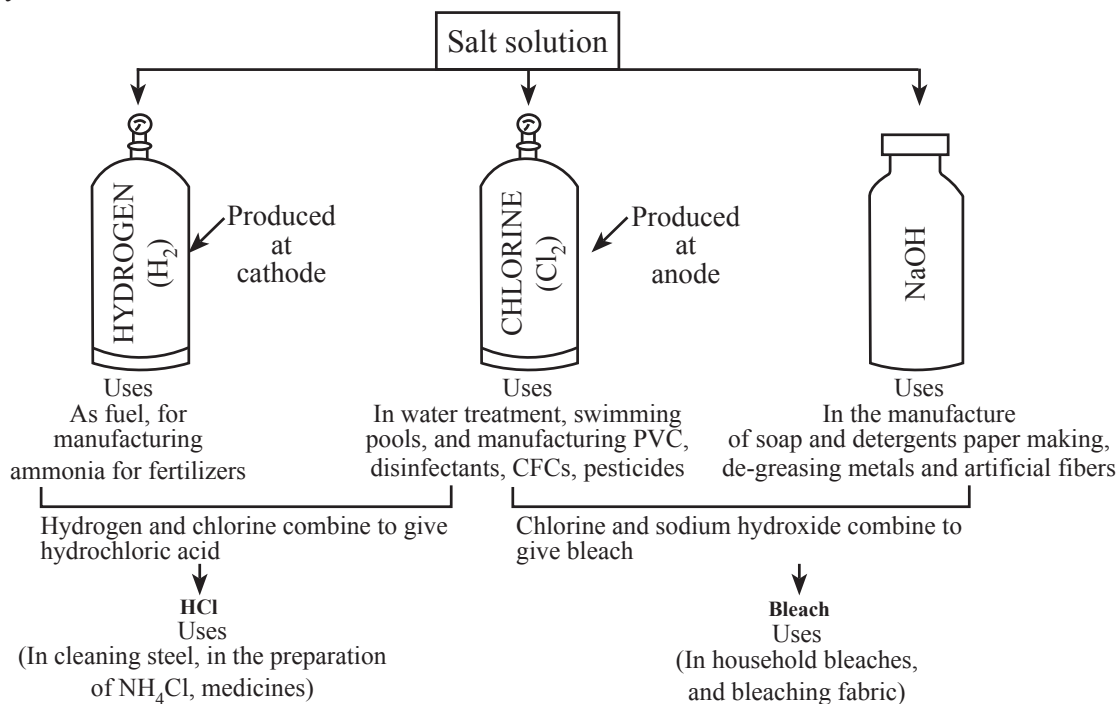


Fig. 3: Chlor-alkali process

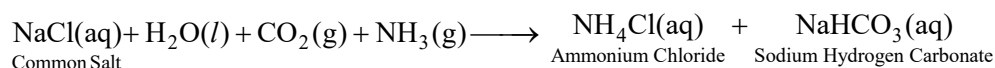
## Washing Soda ( $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ )

**Chemical name:** Sodium carbonate decahydrate

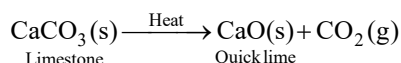
**Chemical formula:**  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

- ❑ Washing soda is manufactured by Solvay's Process.
- ❑ The raw materials that are needed for this process are sodium chloride, limestone ( $\text{CaCO}_3$ ) and ammonia ( $\text{NH}_3$ ). The reactions involved are:

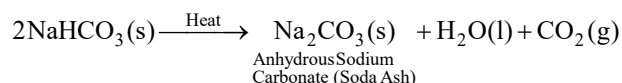
**First step:**



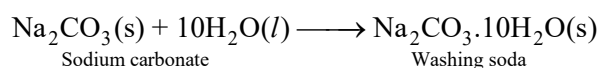
The  $\text{CO}_2$  that is required in this reaction is obtained by heating limestone.



**Second step:** Dry sodium hydrogen carbonate is heated strongly to give sodium carbonate



**Third step:** Sodium carbonate is recrystallized by dissolving in water to get washing soda. It is a basic salt.

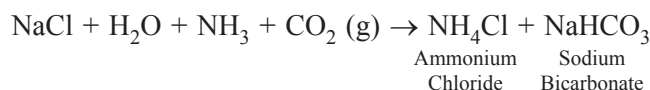


## Uses of Washing Soda

- ❑ It is used as clearing agent for domestic purpose.
- ❑ It is used for removing permanent hardness of water.
- ❑ It is used in glass, soap and paper industries.
- ❑ It is used in manufacture of borax ( $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ ).

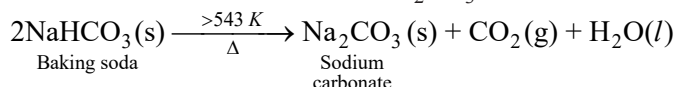
## Baking Soda ( $\text{NaHCO}_3$ )

- ❑ Baking soda is chemically known as **sodium hydrogen carbonate** or **sodium bicarbonate** ( $\text{NaHCO}_3$ ).
- ❑ Baking soda is obtained as an intermediate product in the preparation of  $\text{Na}_2\text{CO}_3$  by solvay process. In this process, a saturated sodium chloride solution in water is saturated with  $\text{NH}_3$  and then carbon dioxide gas is passed into the liquid. Sodium chloride, ( $\text{NaCl}$ ) is converted into sodium bicarbonate ( $\text{NaHCO}_3$ ) which being less soluble, separates out from the solution.



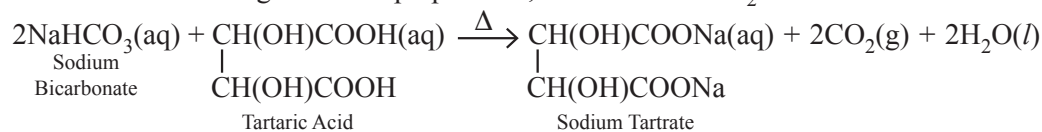
## Properties of $\text{NaHCO}_3$

1. It is a white crystalline substance which forms an alkaline solution with water. The aqueous solution of  $\text{NaHCO}_3$  changes the colour of methyl orange to yellow but gives pink colour with phenolphthalein.
2. When  $\text{NaHCO}_3$  is heated above 543 K, it is converted into  $\text{Na}_2\text{CO}_3$ .



## Uses of Baking Soda

1. It is used in the manufacture of baking powder. Baking powder is a mixture of baking soda and a mild edible acid such as tartaric acid or citric acid. During the bread preparation, the release of  $\text{CO}_2$  causes the bread to rise (swell).



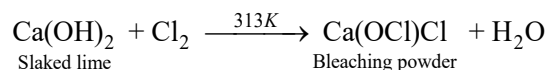
2. It acts as an antacid.
3. Used in a soda-acid fire extinguishers.
4. It used in the tanning, textile, and paper industries.

## Bleaching Powder ( $\text{CaOCl}_2$ )

- Bleaching powder is commercially called 'chloride of lime or' chlorinated lime'.

### Preparation

- Bleaching powder is prepared by passing chlorine gas over dry slaked lime at 313 K.



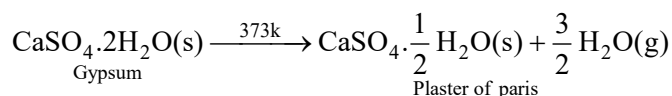
### Uses

1. It is used as a bleaching agent in paper and textile industries.
2. Used to prepare chloroform ( $\text{CHCl}_3$ ).
3. It is used for disinfecting water.
4. It is also used to make wool shrink-proof.
5. It is used as an oxidising agent in many chemical industries.

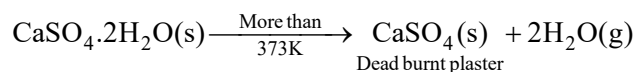
## Plaster of Paris ( $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ )

**Chemical Name:** Calcium Sulphate Hemihydrate

### Preparation



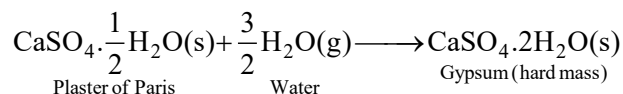
During the preparation of plaster of Paris (POP), temperature is to be controlled carefully. Else, anhydrous calcium sulphate ( $\text{CaSO}_4$ ) formation take place. Anhydrous calcium sulphate does not set into hard mass when it is mixed with water. So, if temperature is not set, then Plaster of Paris formed will have poor setting property.



### Properties of POP

1. At ordinary room temperature, Plaster of Paris absorbs water and a large amount of heat is released.
2. When mixed with a limited amount of water (50% by mass), then it will forms a plastic mass, which release heat and quickly sets to a hard porous mass within minutes. This is known as **setting process**.

During setting, there is a slight expansion in volume occurs. It is due to this that it will fills the mould completely and gives sharp impression. The reaction during process is



## Uses of POP

1. It is also used for making toys, cosmetic and casting statues.
2. It is used in the laboratories for sealing gaps where airtight arrangement is needed.
3. It is used in making moulds in pottery.

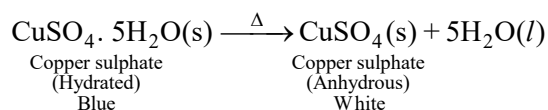
## Water of Crystallisation

Certain salts contain a fixed amount of some  $\text{H}_2\text{O}$  molecules loosely attached to their own molecules. They are known as hydrated salts and they are of crystalline nature. The molecules of  $\text{H}_2\text{O}$  present in one formula unit of salt is called as 'water of crystallisation'.

**Hydrated salts:** There are certain salts in which water of crystallisation is present. These salts are known as hydrated salts. E.g., sodium carbonate decahydrate is  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ , calcium sulphate dihydrate  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ .

**Anhydrous salts:** There are certain salts in which water of crystallisation is absent. These salts are known as anhydrous salts. E.g., Anhydrous copper sulphate, anhydrous ferrous sulphate.

The molecules of water of crystallisation also account for their specific colours in coloured crystalline and hydrated salts. Hence, upon heating of hydrated salt, its colour changes as molecules of water of crystallisation are removed and the salt becomes anhydrous. For example, if we take a few crystals of blue vitriol i.e. hydrated copper sulphate in a dry or boiling tube then the salt will change colour to a white anhydrous powder and water droplets will appear on the walls of the tube. Further, the tube is cooled and few drops of water are added again. The white anhydrous powder will again gain blue colour.



## Activity

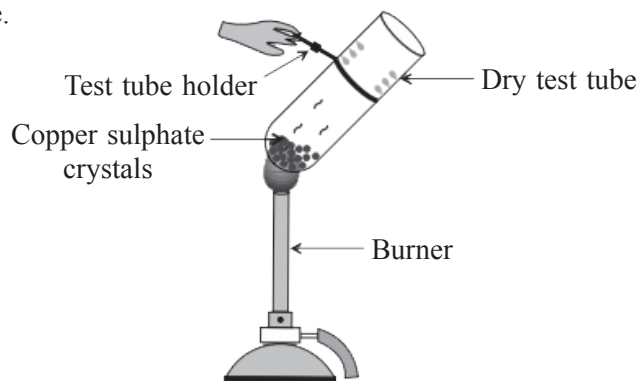
**Aim:** To test the presence of water of crystallisation in copper sulphate crystals.

**Procedure:**

- ❑ Heat a few crystals of copper sulphate in a dry boiling tube.
- ❑ Note the colour of copper sulphate after heating.
- ❑ Observe the water droplets in the boiling tube.
- ❑ Add 2-3 drops of water on the sample of copper sulphate obtained after heating.

**Observation:** Blue coloured copper sulphate crystals on heating leave behind white anhydrous copper sulphate and water droplets are seen in the upper cooler parts of the boiling tube. On adding 2-3 drops of water to the white residue, blue colour reappears.

**Conclusion:** Copper sulphate crystals on heating lose water to form white anhydrous copper sulphate which combines with water to form blue coloured copper sulphate crystals.





## Summary

### Indicators

- ❖ An indicator is a chemical compound which indicates the presence of acidic, basic or neutral substance either by change in colour or odour. for e.g. litmus solution.
- ✦ **Natural indicators:** Turmeric, litmus (obtained from lichen), china rose, red cabbage
- ✦ **Synthetic indicators:** Methyl orange, phenolphthalein.
- ✦ **Olfactory indicators:** Those substances whose odour changes in acidic or basic media are known as Olfactory indicators. For e.g., clove oil, vanilla essence, onion.

| Indicator         | Acid solution | Basic solution | Neutral solution |
|-------------------|---------------|----------------|------------------|
| Red litmus        | No change     | Blue           | No change        |
| Blue litmus       | Red           | No change      | No change        |
| Turmeric          | Yellow        | Reddish brown  | Yellow           |
| Red cabbage juice | Red           | Green          | Purple           |
| Phenolphthalein   | Colourless    | Pink           | Colourless       |
| Methyl Orange     | Red           | Yellow         | Orange           |

### Acids

- ❖ **Acids** are those substances which release one or more  $H^+$  ions in aqueous solution.
- Example:** Sulphuric acid ( $H_2SO_4$ ), Acetic Acid ( $CH_3COOH$ ) etc.
- ✦ Acidic solution conducts electricity.
- ✦ Acids are sour in taste and turn blue litmus red.

### Bases

- ❖ **Bases** are those substances which release hydroxide ions ( $OH^-$ ) in aqueous solution.
- Examples:** Sodium hydroxide (caustic soda) —  $NaOH$ ,  $Ca(OH)_2$  etc.
- ✦ Bases are bitter in taste, have soapy touch and turn red litmus blue.

### Chemical Reaction of Acids and Bases

- Acid + active metal  $\rightarrow$  metal salt + hydrogen gas  
Base + amphoteric metal  $\rightarrow$  salt + hydrogen gas
- Acid + metal carbonate  $\rightarrow$  metal salt + water + carbon dioxide gas  
Base + metal carbonate  $\rightarrow$  No reaction
- Acid + metal bicarbonate  $\rightarrow$  metal salt + water + carbon dioxide gas  
Base + metal bicarbonate  $\rightarrow$  No reaction
- Acid + Base  $\rightarrow$  salt + water
- Metal oxide are basic in nature  
Acid + metal oxide  $\rightarrow$  Salt + Water  
Non-metal oxides are acidic in nature  
Base + Non-metal oxide  $\rightarrow$  Salt + Water

## pH Scale

Strength of acid and base is expressed in terms of pH scale that ranges from 0 to 14 using pH paper.

Negative logarithm of hydrogen ion concentration is known as pH.

*i.e.*,  $\text{pH} = -\log[\text{H}^+]$

In pH scale, Acids: 0 to less than 7 ; bases: more than 7 to 14 ; neutral: 7

## Salts

A **salt** is a compound formed by the reaction of an acid with a base.

## Chemicals from Common Salt

Sodium chloride is called as common salt and is used in our food. It is extracted from seawater.

Following is a list of common chemical that are used in our daily life:

### 1. Sodium hydroxide (NaOH):

**Preparation:**  $2\text{NaCl}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \xrightarrow{\text{Electric current}} 2\text{NaOH}(\text{aq}) + \text{Cl}_2(\text{g}) + \text{H}_2(\text{g})$

**Uses:** It is used to make soap and as a main ingredient in household products such as liquid drain cleaners.

### 2. Bleaching Powder (CaOCl<sub>2</sub>):

**Preparation:**  $\text{Ca}(\text{OH})_2(\text{aq}) + \text{Cl}_2(\text{g}) \xrightarrow{313\text{ K}} \text{Ca}(\text{OCl})\text{Cl}(\text{s}) + \text{H}_2\text{O}(\text{g})$

**Uses:** Used in textile factories, laundries and as a disinfectant.

### 3. Baking soda (NaHCO<sub>3</sub>):

**Preparation:**  $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$

**Uses:** Used in glass manufacturing, soap and paper industry, as cleaning agent for domestic purpose, to make baking powder, antacids etc.

### 4. Washing soda (Na<sub>2</sub>CO<sub>3</sub>·10H<sub>2</sub>O):

**Preparation:**  $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$

Dry Sodium hydrogen carbonate is heated strongly to produce sodium carbonate.

$2\text{NaHCO}_3(\text{s}) \xrightarrow{\text{Heat}} \underset{\substack{\text{Anhydrous Sodium} \\ \text{Carbonate (Soda Ash)}}}{\text{Na}_2\text{CO}_3(\text{s})} + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$

$\underset{\substack{\text{Sodium} \\ \text{carbonate}}}{\text{Na}_2\text{CO}_3(\text{s})} + 10\text{H}_2\text{O}(\text{l}) \longrightarrow \underset{\substack{\text{Washing soda}}}{\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}(\text{s})}$

**Uses:** For softening of hard water, used in detergents as it has cleansing properties.

### 5. Plaster of Paris (CaSO<sub>4</sub>· $\frac{1}{2}$ H<sub>2</sub>O):

On heating gypsum at 373K. It becomes CaSO<sub>4</sub>· $\frac{1}{2}$ H<sub>2</sub>O that is plaster of Paris

$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(\text{s}) \longrightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}(\text{s}) + \frac{3}{2}\text{H}_2\text{O}(\text{g})$

If temperature is more than 373 K.

$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(\text{s}) \xrightarrow[373\text{ K}]{\text{More than}} \underset{\substack{\text{Dead burnt plaster}}}{\text{CaSO}_4(\text{s})} + 2\text{H}_2\text{O}(\text{g})$

**Uses:** Making toys, decorative materials and smooth surfaces.



### In-text Questions

1. You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. If you are given only red litmus paper. How will you identify the contents of each test tube?

**Ans.** The content in each of the test tubes can be identified by using red litmus paper. The colour change of the red litmus paper can be noticed for the identification.

On litmus paper, the three solutions in the test tubes are poured separately.

The solution that turns red litmus to blue contains a basic solution.

The formed blue litmus paper are divided into two parts.

The solution from the test tube that turns blue litmus paper to red will contain the acidic solution. Solution of the test tube that do not change to either red or blue litmus paper contain water.

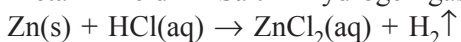
2. Why should curd and sour substances not be kept in brass and copper vessels?

**Ans.** Curd and sour food items contain acids which when combine with metals produce  $H_2$ . Such reaction makes food poisonous that can damage people's health.

3. Which gas is usually liberated when an acid reacts with a metal? Illustrate with an example. How will you test for the presence of this gas?

**Ans.** When an acid reacts with any metal, salt and hydrogen gas are formed.

Metal + Acid  $\rightarrow$  Salt + Hydrogen gas



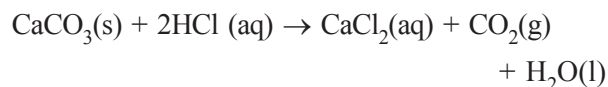
If we bring candle/matchstick near  $H_2$  gas it will burn with a popping sound.

4. Metal compound A reacts with dilute hydrochloric acid to produce effervescence. The gas evolved extinguishes a burning candle. Write a balanced chemical equation for the reaction if one of the compounds formed is calcium chloride.

**Ans.** Since the gas evolved extinguishes a burning candles so the gas released here is  $CO_2$ .

Metal compound A will be calcium carbonate as there is release of  $CO_2$  and one of the compound form is calcium chloride.

Hence the reaction between Calcium Carbonate and HCl is



5. Why do HCl,  $HNO_3$ , etc., show acidic characters in aqueous solutions while solutions of compounds like alcohol and glucose do not show acidic character?

**Ans.** The dissociation of HCl or  $HNO_3$  to form hydrogen ions always occurs in the presence of water hydrogen ions ( $H^+$ ) combine with  $H_2O$  to form hydronium ions ( $H_3O^+$ ). Although aqueous solutions of glucose and alcohol contain hydrogen, these cannot dissociate in water to form hydrogen ion. Hence, they do not show acidic character.

6. Why does an aqueous solution of an acid conduct electricity?

**Ans.** Charged particles are mainly responsible for the conductance of electricity in an acid. These charged particles are known as ions which helps in conductance of electricity in an acid.

7. Why does dry HCl gas not change the colour of the dry litmus paper?

**Ans.** HCl gas will not give  $H^+$  ions in dry state thus HCl does not show any acidic behaviour and colour of the litmus paper will remain the same on reacting with HCl gas.

8. While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid?

**Ans.** The process of dissolving an acid in water is a highly exothermic, The acid must always be added slowly to water with constant stirring. If water is added to a concentrated acid, the heat generated may cause the mixture to splash out and causes burns. The glass container may also break due to excessive heating.

9. How is the concentration of hydronium ions ( $\text{H}_3\text{O}^+$ ) affected when a solution of an acid is diluted?

**Ans.** When acid is added to water there will be a definite amount of  $\text{H}_3\text{O}^+$  present in the fixed volume of solution. If we add water, hydronium ion per volume of solution decrease, this in-turn decreases  $\text{H}_3\text{O}^+$  concentration in the solution.

10. How is the concentration of hydroxide ions ( $\text{OH}^-$ ) affected when excess base is dissolved in a solution of sodium hydroxide?

**Ans.** On dissolving excess base in a solution of sodium hydroxide, concentration of  $\text{OH}^-$  ions per unit volume in the solution increase.

11. You have two solutions, A and B. The pH of solution A is 6 and pH of solution B is 8. Which solution has more hydrogen ion concentration? Which of this is acidic and which one is basic?

**Ans.** A pH value of less than 7 indicates an acidic solution, while greater than 7 indicates a basic solution. Therefore, the solution A with  $\text{pH} = 6$  is acidic and has more hydrogen ion concentration than the solution of  $\text{pH} = 8$  which is basic.

12. What effect does the concentration of  $\text{H}^+(\text{aq})$  ions have on the nature of the solution?

**Ans.** Hydrogen ion concentration decides the nature of the solution. If  $\text{H}^+$  concentration increase then solution will turn acidic and similarly if  $\text{H}^+$  concentration decreases then solution will turn basic.

13. Do basic solutions also have  $\text{H}^+(\text{aq})$  ions? If yes, then why are these basic?

**Ans.** Yes, basic solution also has  $\text{H}^+$  ions in aqueous solution. However, their concentration is less as

compared to the concentration of  $\text{OH}^-$  ions that makes the solution basic.

14. Under what soil condition do you think a farmer would treat the soil of his fields with quick lime (calcium oxide) or slaked lime (calcium hydroxide) or chalk (calcium carbonate)?

**Ans.** If the soil is acidic in nature (pH is below 7) then in such case field will be treated with quick lime (calcium oxide) or slaked lime (calcium hydroxide) or chalk (calcium carbonate).

15. What is the common name of the compound  $\text{CaOCl}_2$ ?

**Ans.** Common name of  $\text{CaOCl}_2$  is bleaching powder.

16. Name the substance which on treatment with chlorine yields bleaching powder

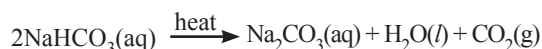
**Ans.** The substance that on treatment with chlorine yields bleaching powder is Calcium hydroxide.

17. Name the sodium compound which is used for softening hard water.

**Ans.** Washing soda ( $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ ) is the compound that is used for softening hard water.

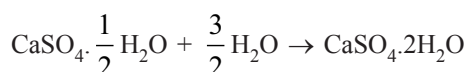
18. What will happen if a solution of sodium hydrogen carbonate is heated? Give the equation of the reaction involved.

**Ans.** Sodium hydrogen carbonate on heating yields sodium carbonate, carbon dioxide gas and water is released in the process.



19. Write an equation to show the reaction between Plaster of Paris and water.

**Ans.** The chemical equation for the reaction of Plaster of Paris and water is



## NCERT Exercise



1. A solution turns red litmus blue, its pH is likely to be

- (a) 1                                      (b) 4  
(c) 5                                      (d) 10

**Ans. (d)** Red litmus paper turns blue in basic solution which has pH more than 7. Thus pH 10 is the correct answer.

2. A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains

- (a) NaCl                                      (b) HCl  
(c) LiCl                                      (d) KCl

**Ans. (b)** Answer is HCl. Egg shells contains calcium carbonate, that react with HCl to release  $\text{CO}_2$  gas that turn lime water to milky.



3. 10mL of a solution of NaOH is found to be completely neutralised by 8mL of a given solution of HCl. If we take 20mL of the same solution of NaOH, the amount of HCl solution (the same solution as before) required to neutralise it will be

- (a) 4 mL                                      (b) 8 mL  
(c) 12 mL                                      (d) 16 mL

**Ans. (d)** As 10ml of NaOH requires 8mL of HCl solution to neutralise.

$$\therefore 20\text{ml of NaOH require } 8 \times 2 = 16\text{mL of HCl}$$

Thus, the answer is option (d).

4. Which one of the following types of medicines is used for treating indigestion?

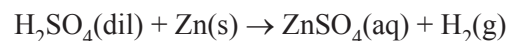
- (a) Antibiotic                                      (b) Analgesic  
(c) Antacid                                      (d) Antiseptic

**Ans. (c)** Excess production of acid in the stomach leads to indigestion. Medicines that is used to treat indigestion is known as Antacid.

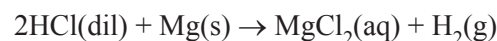
5. Write word equations and then balanced equations for the reaction taking place when

- (a) Dilute sulphuric acid reacts with zinc granules.  
(b) Dilute hydrochloric acid reacts with magnesium ribbon.  
(c) Dilute sulphuric acid reacts with aluminium powder.  
(d) Dilute hydrochloric acid reacts with iron filings.

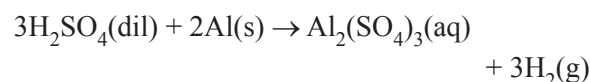
**Ans. (a)** Dilute sulphuric acid + Zinc  $\rightarrow$  Zinc Sulphate + Hydrogen Gas



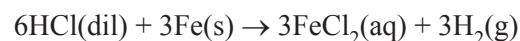
(b) Dilute Hydrochloric acid + Magnesium  $\rightarrow$  Magnesium Chloride + Hydrogen Gas



(c) Dilute Sulphuric Acid + Aluminium  $\rightarrow$  Aluminium Sulphate + Hydrogen Gas

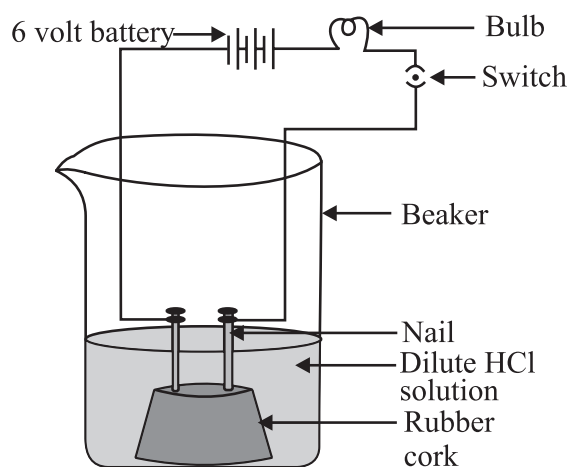


(d) Dilute Hydrochloric Acid + Iron  $\rightarrow$  Ferrous Chloride + Hydrogen Gas



6. Compounds such as alcohols and glucose also contain hydrogen but are not categorised as acids. Describe an activity to prove it.

**Ans.** Two nails are inserted on the wooden or rubber cork and placed them on a beaker as shown in figure below. Iron nail is connected to a bulb, then 6 volt battery and a wire is connected to a switch. Some alcohol or glucose is poured to dip the nails in the glucose or alcohol solution. The switch is turned on and we will see the bulb is not glowing despite being connected to switch. The beaker is emptied then and HCl solution is added. This time bulb glows. This proves that acid can conduct electricity but alcohol and glucose will not conduct electricity.



7. Why does distilled water not conduct electricity, whereas rain water does?

**Ans.** Distilled water does not conduct electricity because it does not contain any ionic compound (like acids, bases or salts) dissolved in it.

Rainwater will dissolved acidic gas such as  $\text{CO}_2$  from the air and  $\text{CO}_2$  gas will forms carbonic acid. This means that it has  $\text{H}^+$  ions and carbonate ions. Thus, with the presence of acids, rainwater will conduct electricity.

8. Why do acids not show acidic behaviour in the absence of water?

**Ans.** The acidic nature from acids is due to the presence of  $H^+$  ions. Hydrogen ions is produced in the presence of water and thus water is needed, if acids wanted to show their acidic behaviour.

9. Five solutions A, B, C, D and E when tested with universal indicator showed pH as 4, 1, 11, 7 and 9, respectively. Which solution is

- (a) Neutral?                      (b) Strongly alkaline?  
(c) Strongly acidic?            (d) Weakly acidic?  
(e) Weakly alkaline?

**Ans.** In increasing order of hydrogen ion concentration:

- (a) Neutral  $\rightarrow$  Solution D with pH 7.
- (b) Strongly alkaline  $\rightarrow$  Solution C with pH 11.
- (c) Strongly acidic  $\rightarrow$  Solution B with pH 1.
- (d) Weakly acidic  $\rightarrow$  Solution A with pH 4.
- (e) Weakly alkaline  $\rightarrow$  Solution E with pH 9.

The pH can be arranged in the increasing order of the concentration of hydrogen ions are:  
 $11 < 9 < 7 < 4 < 1$ .

10. Equal lengths of magnesium ribbons are taken in test tubes A and B. Hydrochloric acid (HCl) is added to test tube A, while acetic acid ( $\text{CH}_3\text{COOH}$ ) is added to test tube B. Amount and concentration taken for both the acids are same. In which test tube will the fizzing occur more vigorously and why?

**Ans.** Hydrochloric acid (HCl) is a strong acid while acetic acid ( $\text{CH}_3\text{COOH}$ ) is a weak acid. The reaction of the acid on the magnesium ribbon will result in formation of  $\text{H}_2$ . Since HCl is a very strong acid there is a lot of evolution of  $\text{H}_2$  gas from test tube A which result in fizzing thus, more fizzing take place in test tube A.

**11.** Fresh milk has a pH of 6. How do you think the pH will change as it turns into curd? Explain your answer.

**Ans.** Fresh milk will change into curd due to formation of lactic acid. Lactic acid will reduces the pH of the milk.

12. A milkman adds a very small amount of baking soda to fresh milk.

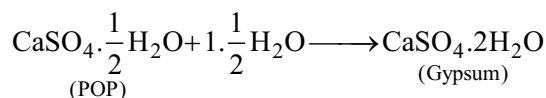
- Why does he shift the pH of the fresh milk from 6 to slightly alkaline?
- Why does this milk take a long time to set as curd?

**Ans.** (a) Milk man will shift the pH of the fresh milk from 6 to slightly alkaline to avoid milk from getting sour due to formation of lactic acid.

(b) This milk will take long time to set into curd because the lactic acid produced here will first neutralises the pH then the pH is reduced to turn milk into curd.

**13.** Plaster of Paris should be stored in a moisture-proof container. Explain why?

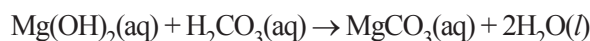
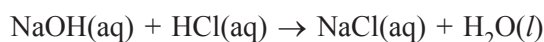
**Ans.** Plaster of Paris should be stored in moisture-proof container because plaster of paris, a powdery mass, adsorbs water (moisture) to form a hard solid known as gypsum.



**14.** What is a neutralisation reaction? Give two examples.

**Ans.** A reaction in which an acid and base react with each other to give a salt and water is known as neutralisation reaction.

### Examples:



**15.** Give two important uses of washing soda and baking soda.

**Ans. Washing soda:**

1. It is used in glass, soap and paper industries.
2. It is used to remove permanent hardness of water.

**Baking soda:**

1. It acts as an antacid.
2. It is used in soda acid fire extinguishers.



## Quick Recall

### Fill in the Blanks

1. Acids turn \_\_\_\_\_ litmus to \_\_\_\_\_.
2. Basic solution of pH is always \_\_\_\_\_ than 7.
3.  $\text{KNO}_3$  has pH value equal to \_\_\_\_\_.
4. The presence of \_\_\_\_\_ in acids is responsible for their acidic properties.
5. Mild base like \_\_\_\_\_ gives relief on the bee-stung area.
6. \_\_\_\_\_ is the fixed number of water molecules that is attached chemically to each formula unit of a salt in its crystalline form.
7. The stomach produces too much \_\_\_\_\_ during indigestion and this causes pain and irritation.
8. \_\_\_\_\_, are the products formed when bleaching powder reacts with dilute sulphuric acid.
9. The chemical formula of Washing soda is \_\_\_\_\_.
10.  $\text{CH}_3\text{COONa}$  salt is \_\_\_\_\_ in nature.

### True and False Statements

1. Litmus (indicator) is obtained from nettle plant.
2. Methyl orange indicator gives yellow colour in basic solution.
3. The strength of an acid or base is measured on the basis of scale of numbers known as pH paper.
4. The more alkaline a solution, the more is the pH.
5. Glucose solution will conduct electricity.
6. The sodium chloride aqueous solution is acidic in nature.
7. Oxalic acid is present in ant sting.
8. The dilution of a concentrated acid is always done by adding concentrated acid to water slowly slowly with stirring.
9. Plaster of Paris can be stored in any type of containers.
10. Carbonic acid is a weak acid.

### Match the Followings

In this section, each question has two matching lists. Choices for the correct combination from Column-I and Column-II are given as options (a), (b), (c) and (d) out of which one is correct.

1. Match the Column-I with Column-II.

| Column-I<br>pH value |            | Column-II<br>Solution |             |
|----------------------|------------|-----------------------|-------------|
| P.                   | 6.5 to 6.7 | (i)                   | Vinegar     |
| Q.                   | 7.4        | (ii)                  | Milk        |
| R.                   | 2.4 – 3.4  | (iii)                 | Human blood |
| S.                   | 12.4       | (iv)                  | Lime water  |

- (a) P-(i), Q-(i), R-(ii), S-(iii)  
 (b) P-(i), Q-(ii), R-(iii), S-(iv)  
 (c) P-(ii), Q-(iii), R-(i), S-(iv)  
 (d) P-(iii), Q-(iv), R-(i), S-(ii)

2. Match the Column-I with Column-II.

| Column-I |                        | Column-II |                        |
|----------|------------------------|-----------|------------------------|
| P.       | Water is produced      | (i)       | Metal + acid           |
| Q.       | $\text{H}_3\text{O}^+$ | (ii)      | Acid + Base            |
| R.       | $\text{H}_2$           | (iii)     | Metal carbonate + acid |
| S.       | Carbon dioxide gas     | (iv)      | Acid + water           |

- (a) P-(i), Q-(iv), R-(iii), S-(ii)  
 (b) P-(ii), Q-(iv), R-(i), S-(iii)  
 (c) P-(ii), Q-(i), R-(iii), S-(iv)  
 (d) P-(i), Q-(iii), R-(iv), S-(ii)

3. Match the Column-I with Column-II.

| Column-I |                              | Column-II |                          |
|----------|------------------------------|-----------|--------------------------|
| P.       | Manufacture of antacid       | (i)       | $\text{NaOH}$            |
| Q.       | Preservation of food         | (ii)      | $\text{H}_2\text{SO}_4$  |
| R.       | Manufacturing of soap        | (iii)     | $\text{NaHCO}_3$         |
| S.       | Used in Automobile batteries | (iv)      | $\text{CH}_3\text{COOH}$ |

- (a) P-(iii), Q-(iv), R-(i), S-(ii)  
 (b) P-(ii), Q-(iv), R-(i), S-(iii)  
 (c) P-(i), Q-(iv), R-(iii), S-(ii)  
 (d) P-(ii), Q-(iii), R-(i), S-(iv)

4. Match the Column-I with Column-II.

| Column-I |                                                      | Column-II |                        |
|----------|------------------------------------------------------|-----------|------------------------|
| P.       | $\text{Mg}(\text{OH})_2$                             | (i)       | Plaster of Paris (POP) |
| Q.       | $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$ | (ii)      | Gypsum                 |
| R.       | $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$            | (iii)     | Bleaching Powder       |
| S.       | $\text{CaOCl}_2$                                     | (iv)      | Milk of magnesia       |

(a) P-(iv), Q-(i), R-(ii), S-(iii)

(b) P-(iv), Q-(ii), R-(iii), S-(i)

(c) P-(i), Q-(iv), R-(iii), S-(ii)

(d) P-(ii), Q-(iii), R-(i), S-(iv)

5. Match the Column-I with Column-II.

| Column-I |                  | Column-II |                                              |
|----------|------------------|-----------|----------------------------------------------|
| P.       | Bleaching powder | (i)       | Preparation of glass                         |
| Q.       | Baking soda      | (ii)      | Production of $\text{H}_2$ and $\text{Cl}_2$ |
| R.       | Washing soda     | (iii)     | Decolourisation                              |
| S.       | Sodium chloride  | (iv)      | Antacid                                      |

(a) P-(ii), Q-(i), R-(iv), S-(iii)

(b) P-(iii), Q-(ii), R-(iv), S-(i)

(c) P-(iii), Q-(iv), R-(i), S-(ii)

(d) P-(ii), Q-(iv), R-(i), S-(iii)

## Assertion & Reason Type Questions

**Direction:** In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice.

- (a) Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is the correct explanation of **Assertion (A)**.
- (b) Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is not the correct explanation of **Assertion (A)**.
- (c) **Assertion (A)** is true but **Reason (R)** is false.
- (d) **Assertion (A)** is false but **Reason (R)** is true.

1. **Assertion (A):** Gas bubbles are observed when sodium carbonate is added to dilute hydrochloric acid.

**Reason (R):** Carbon dioxide is given off in the reaction.

2. **Assertion (A):** Ammonia is a base.

**Reason (R):** It does not contain  $\text{OH}^-$  ions.

3. **Assertion (A):** Ammonia Solution is an alkali.

**Reason (R):** Ammonia solution turns blue litmus paper red.

4. **Assertion (A):** Methanoic acid is present in Ant's sting.

**Reason (R):** The body part where ant bite is treated with dry baking soda.

5. **Assertion (A):** Concentrated  $\text{H}_2\text{SO}_4$  can be diluted by adding water drop wise to acid.

**Reason (R):** Concentrated  $\text{H}_2\text{SO}_4$  has a more affinity for water.

## Statement Type Questions

**Direction:** These questions consist of two statements each, printed as Statement-I and Statement-II. While answering these questions, you are required to choose any one of the following four responses.

- (a) Both **Statement-I** and **Statement-II** are correct.
- (b) Both **Statement-I** and **Statement-II** are incorrect.
- (c) **Statement-I** is correct & **Statement-II** is incorrect.
- (d) **Statement-I** is incorrect & **Statement-II** is correct.

1. **Statement-I:** All salts form neutral solutions with  $\text{pH} = 7$ .

**Statement-II:** Salts are formed by the neutralisation of an acid with a base.

2. **Statement-I:** Bleaching powder acts by liberating chlorine.

**Statement-II:**  $\text{Na}_2\text{CO}_3$  is used to remove the hardness of the water.

3. **Statement-I:** All metal oxides are basic in nature.

**Statement-II:** Lemon juice is more acidic than tomato juice.

4. **Statement-I:** The aqueous solution of glucose and alcohol do not show acidic character.

**Statement-II:** Aqueous solution of glucose and alcohol gives  $\text{H}^+$  ions.

5. **Statement-I:** Tooth decay starts when the  $\text{pH}$  of the mouth is higher than 5.5.

**Statement-II:** Tooth enamel is made up of calcium hydroxyapatite and is the softest substance in the body.



## Multiple Choice Questions

- Which of the following compounds is neutral to litmus?  
(a)  $\text{NaNO}_3$  (b)  $\text{CuSO}_4$   
(c)  $\text{Mg}(\text{OH})_2$  (d)  $\text{NaHCO}_3$
- Which one of the following can be used as an acid-base indicator by a blind student?  
(a) Turmeric (b) Litmus  
(c) Vanilla essence (d) Methyl orange
- What is product formed when zinc metals granules reacts with sodium hydroxide?  
(a) Sodium zincate and carbon dioxide  
(b) Sodium zincate and hydrogen gas  
(c) Zinc hydroxide and sodium  
(d) Sodium zinc-oxide and hydrogen gas
- Which of the following compound is not a base?  
(a)  $\text{Al}(\text{OH})_3$  (b)  $\text{NaOH}$   
(c)  $\text{KOH}$  (d)  $\text{NaCl}$
- The acid that is used in flavoured drinks is:  
(a) Boric acid (b) Carbonic acid  
(c) Oxalic acid (d) Acetic acid
- Acids and bases are important because of  
(a) Their effect on farmer's crop  
(b) Their use in industry  
(c) Their effects on human health  
(d) All the above are correct.
- The acidic soil that is not good for health of growing plants, is neutralized by  
(a) Magnesium hydroxide ( $\text{Mg}(\text{OH})_2$ )  
(b) Calcium oxide ( $\text{CaO}$ )  
(c) Sodium chloride ( $\text{NaCl}$ )  
(d) Ammonium hydroxide ( $\text{NH}_4\text{OH}$ )
- The correct statements about an aqueous solution of an acid and of a base from the following are:  
(i) Higher the pH, stronger the acid  
(ii) Higher the pH, weaker the acid  
(iii) Lower the pH, stronger the base  
(iv) Lower the pH, weaker the base  
(a) (i) and (iii) (b) (i) and (iv)  
(c) (ii) and (iii) (d) (ii) and (iv)
- \_\_\_\_\_ is called impure sodium chloride.  
(a) Gypsum salt  
(b) Rock salt  
(c) Blue vitrol salt  
(d) Carbonate salt
- Which one is not used as raw material in the preparation of baking soda?  
(a)  $\text{NH}_3$   
(b)  $\text{NaCl}$   
(c)  $\text{CO}_2$   
(d)  $\text{NaOH}$
- $\text{NaHCO}_3$  is formed by which of the following reaction:  
(a)  $\text{NaOH} + \text{H}_2\text{CO}_3$   
(b)  $\text{NaCl} + \text{Na}_2\text{CO}_3$   
(c)  $\text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$   
(d)  $\text{NaCl} + \text{H}_2\text{CO}_3$
- $\text{NaOH}$  is obtained by the electrolysis of  
(a)  $\text{NaCl}(\text{aq})$   
(b)  $\text{Na}_2\text{CO}_3(\text{aq}) + \text{CaCl}_2(\text{aq})$   
(c)  $\text{NaHCO}_3(\text{aq})$   
(d)  $\text{NaCl}$  (molten)
- Bleaching powder is soluble in cold water giving a milky solution because of  
(a) Available  $\text{Cl}_2$   
(b) Lime present in it  
(c) Formation of calcium carbonate  
(d) The absorption of  $\text{CO}_2$  from atmosphere
- The bleaching powder is prepared by passing chlorine gas into  
(a) Slaked lime  
(b) Quick lime  
(c) Lime  
(d) Plaster of paris at  $120^\circ\text{C}$
- Plaster of Paris is formed  
(a) When water is added to calcium sulphate  
(b) When sulphuric acid is added to calcium hydroxide  
(c) When gypsum is heated to a very low temperature  
(d) When gypsum is heated at  $100^\circ\text{C}$ .



## Subjective Questions

### Very Short Answer Type Questions

1. What is the colour of phenolphthalein in milk of magnesia?
2. Dry ammonia gas has no action on litmus paper, but a solution of ammonia in water turns red litmus paper blue. Why is it so?
3. If the pH of a vegetable soup is 6.5. How is the taste of the soup likely to be?
4. What effect will the concentration of  $H^+(aq)$  will have on the acidic nature of the solution?
5. Name two flowers whose colour of petals can be used as indicator.
6. Which acid is present in honey bee sting and hair of nettle leaves?
7. What effect does an increase in concentration of  $H^+$  in a solution have on the pH of solution?
8. When bitter gourd extract is added to red litmus solution. What will happen? Is it acidic or basic?
9. Name the planet that contains yellowish clouds of sulphuric acid?
10. Name an acid-base indicator prepared at home.
11. Name the gas that is evolved when sodium hydrogen carbonate reacts with dilute hydrochloric acid?
12. The pH of a salt used to make tasty and crispy pakoras is 9. Identify the salt and write a chemical equation for its formation. List its two uses.

### Short Answer Type Questions

1. Name the acid present in the following:  
(a) Tomato (b) Tamarind
2. Name the natural occurrence of each of the given acid  
(a) Citric acid  
(b) Oxalic acid  
(c) Lactic acid  
(d) Tartaric acid
3. 20 mL of water and 15mL of sulphuric acid are to be mixed in a beaker  
(a) State the method that should be followed with reason.  
(b) What is this process known as?

4. A solution 'X' gives orange colour when a drop of it falls on pH paper, while another solution 'Y' gives bluish colour when a drop of it falls on pH paper. What is the nature of both the solutions? Determine the pH of solutions 'X' and 'Y'. (CBSE 2019)
5. A teacher provided acetic acid, water, lemon juice, aqueous solution of sodium hydrogen carbonate and sodium hydroxide to students in the school laboratory to determine the pH values of these substances using pH papers. One of the students reported the pH values of the given substances as 3, 12, 4, 8 and 14 respectively. Which one of these values is not correct? Write its correct value stating the reason. (CBSE 2019)
6. What is the colour of  $FeSO_4 \cdot 7H_2O$  crystals? Will there is any colour change upon heating the  $FeSO_4 \cdot 7H_2O$  crystal? Give balanced chemical equation for the changes.
7. Explain how antacid works.
8. A white coloured powder which is used by doctors for supporting fractured bones. (CBSE 2019)  
(a) Write chemical name and formula of the powder.  
(b) How is this white powder prepared?  
(c) When this white powder is mixed with water a hard solid mass is formed. Write balanced chemical equation for the following change.  
(d) Give one use of this white powder.
9. A student dropped few pieces of marble in a dilute hydrochloric acid contained in a test tube. The evolved gas was passed through lime water.  
(a) What change would be observed in lime water?  
(b) Write balanced chemical equation for the change.
10. How is washing soda prepared from sodium carbonate? Give its chemical equation. State the type of this salt. Name the type of hardness of water which can be removed by it? (CBSE 2020)
11. "Sodium hydrogencarbonate is a basic salt". Justify the statement. The common name of above compound.
12. Give reaction involved in baking soda preparation. Write one of baking soda uses.

13. Define water of crystallisation. Give the chemical formula for two compounds as examples. How can it be proved that the water of crystallisation makes a difference in the state and colour of the compounds? (CBSE 2019)

### Long Answer Type Questions

1. A cloth strip dipped in onion juice is used for testing a liquid 'X'. The liquid 'X' changes its odour. Which type of an indicator is onion juice? The liquid 'X' turns blue litmus red. List the observations the liquid 'X' will show on reacting with the following: (CBSE 2020)

- (a) Zinc granules
- (b) Solid sodium carbonate

Write the chemical equations for the reactions involved.

2. (a) Define indicator. Name two indicators obtained from plants.  
(b) Write a balanced chemical equation for the reaction taking place when sodium oxide reacts with water. How will this solution behave towards phenolphthalein and red litmus paper?  
(c) State what happens when sodium hydroxide solution reacts with hydrochloric acid.
3. Explain the reaction of dilute HCl on the following substance with the help of chemical equation:  
(a) Magnesium ribbon  
(b) Sodium hydroxide  
(c) Crushed egg shells
4. State reason for the following statements: (CBSE 2019)  
(a) Tap water conducts electricity whereas distilled water does not.  
(b) Dry hydrogen chloride gas does not turn blue litmus red whereas dilute hydrochloric acid does.  
(c) During summer season, a milk man usually adds a very small amount of baking soda to fresh milk.  
(d) For a dilution of acid, acid is added into water and not water into acid.  
(e) Ammonia is a base but does not contain hydroxyl group.
5. (a) Identify the acid and the base whose combination forms the common salt. Write the formula and chemical name of this salt. Name the source from which it is obtained.  
(b) What is rock salt? Mention its colour and the reason it is due to.

- (c) What happens when an electric current is passed through brine? Write the chemical equation for it. (CBSE 2020)

6. For making cake, baking powder is taken. If at home your mother uses baking soda instead of baking powder in cake, (NCERT Exemplar)

- (a) How will it affect the taste of the cake and why?
- (b) How can baking soda be converted into baking powder?
- (c) What is the role of tartaric acid added to baking soda?

7. A metal carbonate X on reacting with an acid gives a gas which when passed through a solution Y gives the carbonate back. On the other hand, a gas G that is obtained at anode during electrolysis of brine is passed on dry Y, it gives a compound Z, used for disinfecting drinking water. Identity X, Y, G and Z. (NCERT Exemplar)

8. (a) Crystals of a substance changed their blue colour when heating take place in a closed test tube but regained it after sometime when they were allowed to cool down. Name the substance and write its formula and explain the phenomenon that is involved.

- (b) Name the compound whose one formula unit is associated with 10 water molecules. How is it prepared? Give equations of related reactions. Give two uses of the compound.

9. A sulphate salt of Group 2 element of the Periodic Table is a white, soft substance, which can be moulded into different shapes by making its dough. When this compound is left in open for some time, it becomes a solid mass and cannot be used for moulding purposes. Identify the sulphate salt and why does it show such a behaviour? Give the reaction involved.

### Case-Based Type Questions

#### Case Study-I

A solid compound A on heating gives carbon dioxide gas and a solid residue. The residue when mixed with water forms B. On passing excess of  $\text{CO}_2$  through B in water, a clear solution C is obtained.

1. Identify the compound A.

- (a)  $\text{CaCO}_3$
- (b)  $\text{Na}_3\text{PO}_4$
- (c)  $\text{CaO}$
- (d)  $\text{Ca}(\text{OH})_2$

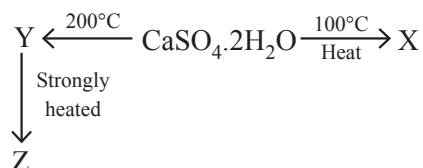
2. Identify the compound B.

- (a)  $\text{CO}_2$  (b)  $\text{CaCO}_3$   
(c)  $\text{Ca(OH)}_2$  (d)  $\text{CaCl}_2$

3. Write the formula of the compound C.

- (a)  $\text{Ca(HCO}_3)_2$  (b)  $\text{Ca(OH)}_2$   
(c)  $\text{CaO}$  (d)  $\text{CaCO}_3$

### Case Study-II



1. Name the compound X:

- (a) Plaster of paris  
(b) Calcium hydroxide  
(c) Slaked lime  
(d) Dead burnt plaster

2. Name the compound Y:

- (a) Dead burnt plaster (b) Limestone  
(c) Lime water (d) Plaster of Paris

3. Name the compound Z:

- (a) Calcium hydroxide (b) Lime  
(c) Dead burnt plaster (d) Plaster of Paris

### Case Study-III

A compound P of sodium forms a white powder and it is constituent of baking powder and is used in some medicine that is used to treat acidity known as antacids. When heated it gives a compound Q that is anhydrous in nature and become hydrated salt when water is absorbed. This salt when kept open in air, loses water molecule in a process known efflorescence.

1. What is formula of Q?

- (a)  $\text{NaHCO}_3$  (b)  $\text{Na}_2\text{CO}_3$   
(c)  $\text{NaOH}$  (d)  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

2. What happens when sodium carbonate decahydrate is exposed to air?

- (a) It loses two molecule of water.  
(b) It will dissociates to give carbon dioxide.  
(c) It loses nine molecule of water.  
(d) It loses ten molecule of water.

3. What is the nature of the solution formed by dissolving Q in water?

- (a) Alkaline (b) It remains insoluble  
(c) Neutral (d) Acidic

4. Identify R.

- (a)  $\text{CO}_2$  (b)  $\text{H}_2\text{CO}_3$   
(c)  $\text{NaOH}$  (d)  $\text{NaCl}$



## Competitive Level

### Classification of Acids on the Basis of their Source

1. Mineral acids
2. Organic acids

#### Mineral Acids

- ❑ Acids that are obtained from rocks and minerals are known as mineral acids. *e.g.* HCl, HNO<sub>3</sub>

**Table-1:** Some of the commonly used mineral acid along with their uses are given below as:

|    | Name              | Type of acid | Chemical Formula               | Uses                                                                                                              |
|----|-------------------|--------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1. | Hydrochloric acid | Mineral Acid | HCl                            | Purification of NaCl, to make aqua regia (mixture of HCl and HNO <sub>3</sub> in ratio of 3 : 1), bleaching agent |
| 2. | Phosphoric acid   | Mineral Acid | H <sub>3</sub> PO <sub>4</sub> | In fertilizers and in antirust paints.                                                                            |
| 3. | Carbonic acid     | Mineral acid | H <sub>2</sub> CO <sub>3</sub> | Used in tanning industry, in soft drinks etc.                                                                     |
| 4. | Sulphuric acid    | Mineral Acid | H <sub>2</sub> SO <sub>4</sub> | Used in car batteries, in paints, plastics, drugs, in petroleum refining.                                         |
| 5. | Nitric acid       | Mineral Acid | HNO <sub>3</sub>               | Used in explosives manufacture like TNT and fertilizers, refining of gold and silver.                             |

#### Organic Acids

- ❑ Acids present in plants and minerals (*i.e.* living organisms) are called organic acids.
- ❑ Organic acids are the acids having the group —COOH and mainly consists of carbon skeleton acid *e.g.* Formic acid.
- ❑ All organic acids contain carbon as one of their constituting elements.
- ❑ They are generally weak acids.

**Table-2:** Some of commonly used organic acids along with their chemical formula and uses are given as:

|    | Name         | Type of acid | Chemical Formula                             | Uses                                                                                           |
|----|--------------|--------------|----------------------------------------------|------------------------------------------------------------------------------------------------|
| 1. | Lactic acid  | Organic Acid | CH <sub>3</sub> CH(OH)COOH                   | Flavouring agent and food preservative.                                                        |
| 2. | Benzoic acid | Organic Acid | C <sub>6</sub> H <sub>5</sub> COOH           | Used in dyes, perfumes and insect repellents, food preservative.                               |
| 3. | Acetic acid  | Organic Acid | CH <sub>3</sub> COOH                         | Used as a solvent in the manufacture of dyes and perfumes, in vinegar, production of plastics. |
| 4. | Formic acid  | Organic Acid | HCOOH                                        | In tanning leather, in medicines for treating disease like gout of joints.                     |
| 5. | Citric acid  | Organic Acid | C <sub>6</sub> H <sub>8</sub> O <sub>7</sub> | Flavouring and preserving agent in soft drink and toffees.                                     |

❑

**Table-3:** Some of the Common Organic Acids and Their Natural Sources are given below:

| Substance that contain acids | Organic acid present in them           |
|------------------------------|----------------------------------------|
| 1. Orange, lemon             | Citric acid, ascorbic acid (vitamin C) |
| 2. Apple                     | Malic acid                             |
| 3. Tamarind (imli), grape    | Tartaric acid                          |
| 4. Olive oil                 | Oleic acid                             |
| 5. Vinegar                   | Acetic acid                            |
| 6. Curd (sour milk)          | Lactic acid                            |
| 7. Urine                     | Uric acid                              |
| 8. Tomato                    | Oxalic acid                            |
| 9. Tea                       | Tannic acid                            |
| 10. Red ants                 | Formic acid                            |
| 11. Green leafy vegetables   | Folic acid                             |
| 12. Amla                     | Ascorbic acid                          |
| 13. Spinach                  | Oxalic acid                            |
| 14. Bees and nettle sting    | Formic acid                            |
| 15. Rancid butter            | Butyric acid                           |

### Classification of Acids on the Basis of their Basicity

- ❑ The basicity of an acid is defined as the number of hydronium ions ( $\text{H}_3\text{O}^+$ ) that can be produced by the complete ionisation of one molecule of that acid in aqueous solution.
- ❑ The acids that contain 1 replaceable hydrogen atom in its molecule is known as **monobasic acids** and its basicity is 1.
- ❑ The acids that contain 2 replaceable hydrogen atoms in their molecules are known as a **dibasic acids** and its basicity is 2.
- ❑ The acids that contains 3 replaceable hydrogen atoms in their molecules are known as **tribasic acids** and its basicity is 3.

**Table-4:** Examples of a few acids with their basicities are given in the table below.

| Acid                                        | Basicity |
|---------------------------------------------|----------|
| Hydrochloric (HCl)                          | 1        |
| Nitric acid ( $\text{HNO}_3$ )              | 1        |
| Sulphuric acid ( $\text{H}_2\text{SO}_4$ )  | 2        |
| Phosphoric acid ( $\text{H}_3\text{PO}_4$ ) | 3        |
| Carbonic acid ( $\text{H}_2\text{CO}_3$ )   | 2        |
| Orthoboric acid ( $\text{H}_3\text{BO}_3$ ) | 1        |



### Knowledge Hub

❖  $\text{H}_3\text{PO}_3$  is an oxyacid of phosphorus which is a dibasic acid, as not all the three hydrogens in  $\text{H}_3\text{PO}_3$  are ionisable.

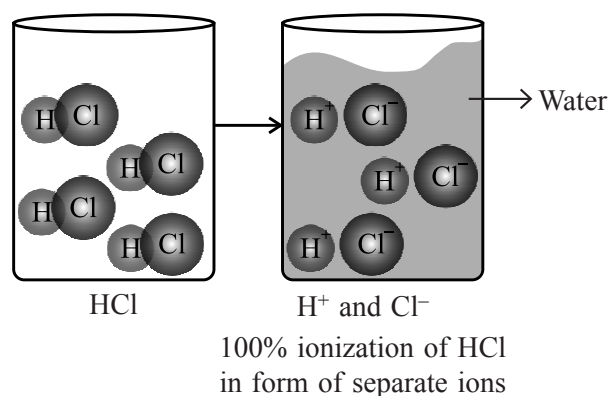
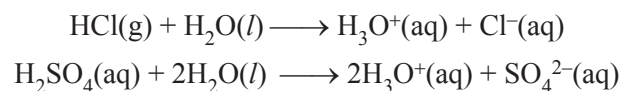
## Classification of Acids on the Basis of their Strength

**1. Strong acids:** Acids that are completely ionised in water are called as strong acids.

- ❑ An aqueous solution of a strong acid contains only ions along with water.

**For Example:**

- ❑ Hydrochloric acid (HCl), sulphuric acid (H<sub>2</sub>SO<sub>4</sub>), nitric acid (HNO<sub>3</sub>) etc. are all considered as strong acids.



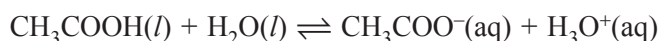
**Fig. 4:** Completely Dissociate of Strong Acids in Water

**2. Weak acids:** Acids that are partially ionised in water are called as weak acids.

- ❑ An aqueous solution of a weak acid contains ions and undissociated molecules along with water.

**For Example:**

- ❑ Carbonic acid (H<sub>2</sub>CO<sub>3</sub>), acetic acid (CH<sub>3</sub>COOH), formic acid (HCOOH), are considered as weak acids.



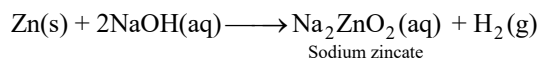
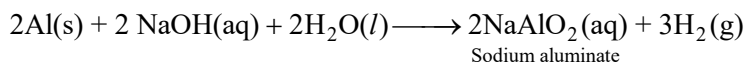
## Classification of Acids on the Basis of their Concentration

- ❑ An acid solution is classified as concentrated or dilute depending upon the amount of the acid that is present in the solution.
- ❑ When the solution contains a larger concentration of the acid, it is known as concentrated acid whereas a dilute solution contains smaller concentration of the acid.

## Chemical Properties of Bases

**1. Reaction of bases with metals**

- ❑ Metals like zinc, tin and aluminum react with strong bases like NaOH, KOH to release hydrogen gas.



**2. Reaction of bases with non-metallic oxides**

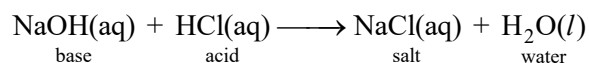
- ❑ Bases react with oxides of non-metals to form salt and water. Non-metal oxides are acidic in nature.

**For Example:**

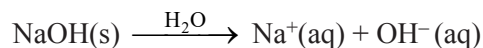


### 3. Reaction of Bases with Acids

- ❑ Bases react with acids to form salt and water.

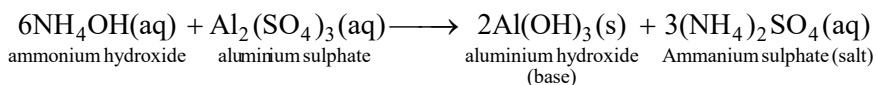


4. The solutions of bases in water give a soapy touch. When bases are dissolved in water they form hydroxide ions ( $\text{OH}^-$ ) in solution.



### 5. Reaction of Bases react with other salts

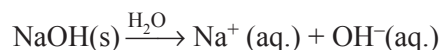
Bases react with certain salts to form another salt and another base. For example, when  $\text{NH}_4\text{OH}$  is added to a solution of  $\text{Al}_2(\text{SO}_4)_3$ ,  $(\text{NH}_4)_2\text{SO}_4$  and  $\text{Al}(\text{OH})_3$  are produced.



## Classification of Bases on the Basis of their Strength

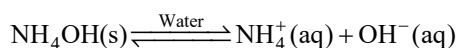
1. **Strong base:** Bases that are completely ionised in water and releases more hydroxyl ( $\text{OH}$ ) groups in an aqueous solution upon ionisation.

**For Example:**  $\text{NaOH}$ , and  $\text{KOH}$  etc.



2. **Weak bases:** Bases that are partially ionised on dissolving in water and has a low concentration of hydroxyl ions.

**For Example:**  $\text{NH}_4\text{OH}$ , and  $\text{Zn}(\text{OH})_2$  etc.



### Knowledge Hub

- ❖ Strong bases can react with the fatty acids and oils that naturally present on your skin's surface. The reaction (known as saponification produces a soap-like substance, which is why bases feels slippery.

## Classification of Bases on the Basis of Their Acidity

The acidity of a base is defined as the number of hydroxyl ( $\text{OH}^-$ ) groups produced by one molecules of an alkali on complete dissociation in water.

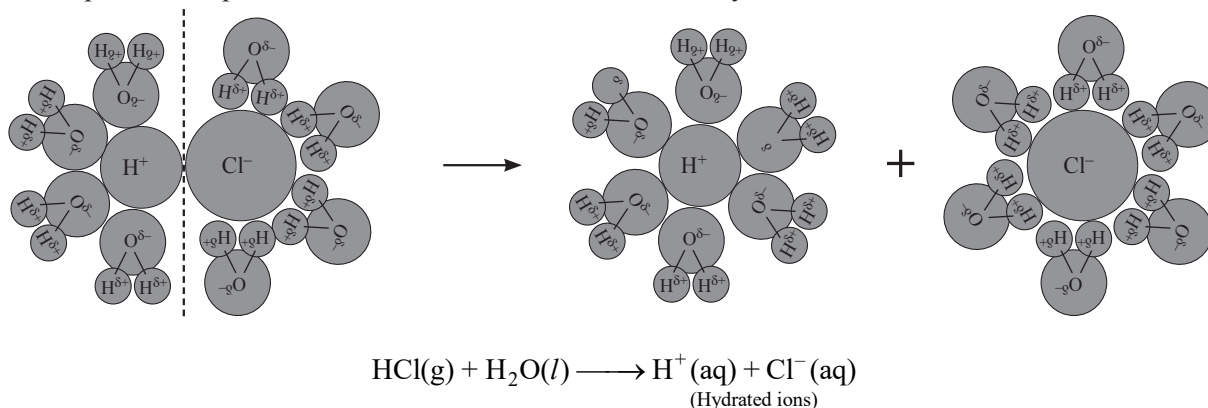
- ❑ The base that contain one hydroxyl group in a molecule is called as **mono acidic base**. *e.g.*,  $\text{NaOH}$ . Its acidity is 1.
- ❑ The base that contain two hydroxyl groups is called as **diacidic base**. *e.g.*,  $\text{Ca}(\text{OH})_2$ . Its acidity is 2.
- ❑ The base that contain three hydroxyl groups is called as **triacidic base**. *e.g.*,  $\text{Fe}(\text{OH})_3$ . Its acidity is 3.

## Comparison Between Acids & Bases

| Acids                                                                                      | Bases                                                                                          |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1. Acids are those that has sour taste.                                                    | 1. Bases are those that have bitter taste.                                                     |
| 2. Acids shows electrolytic conductivity in aqueous solution.                              | 2. Bases shows electrolytic conductivity in aqueous solutions.                                 |
| 3. Acids depends on concentration of $\text{H}^+$ ions.                                    | 3. Bases depends on concentration of $\text{OH}^-$ ions.                                       |
| 4. In acids, litmus paper turns from blue to red, phenolphthalein will remains colourless. | 4. In bases, litmus turns from red to blue, phenolphthalein will turn from colourless to pink. |
| 5. Acids decompose carbonate salts.                                                        | 5. No decomposition of carbonate salts by bases.                                               |

## The Ionisation of Acids & Bases

- Substances can act as acids and bases only in an aqueous solution.
- In dry state, that is, also known as anhydrous state, ionisation of acids & bases cannot be seen.
- Water helps in the ionisation of acids or base by separating the ions.
- This is also known as dissociation and it is explained on the basis of Arrhenius theory of acids and bases.
- In the dry state, Hydrochloric acid (HCl) is known as hydrogen chloride gas i.e.,  $\text{HCl(g)}$ . It will not give any  $\text{H}^+$  ions in dry state. Thus, the acidic character is not seen.
- When HCl gas is passed through water taken in a beaker with the help of glass pipe.  $\text{H}_2\text{O}$  molecules are of polar nature which means that they have partial negative charge ( $\delta^-$ ) on oxygen atom and partial positive charge ( $\delta^+$ ) on hydrogen atoms. They will try to form a sort of envelope around the hydrogen atoms as well as chlorine atoms present in the acid and thus help in their separation as ions. These ions are said to be hydrated ions



- The electrical current is carried through these ions. This applies same for other acids as well as bases. Therefore,
  - Acids are those that release  $\text{H}^+$  ions only in an aqueous solution.
  - Base are those that release  $\text{OH}^-$  ions only in an aqueous solution.
  - Hydration helps in the release of ions from both acids and bases.

### Arrhenius concept

- Arrhenius Acid:** Substance that gives  $\text{H}^+$  ion on dissolving in water ( $\text{H}^+$  donor).

**For Example:**  $\text{HNO}_3$ ,  $\text{HClO}_4$ ,  $\text{HCl}$ ,  $\text{HI}$ ,  $\text{HBr}$ ,  $\text{H}_2\text{SO}_4$ , and  $\text{H}_3\text{PO}_4$  etc.

- Arrhenius base:** Substance that releases  $\text{OH}^-$  (hydroxyl) ion in water ( $\text{OH}^-$  ion donor).

**For Example:**  $\text{NH}_4\text{OH}$ ,  $\text{NaOH}$  and  $\text{KOH}$  etc.

### Bronsted-Lowry concept

- A **Bronsted-Lowry acid** is any species that is capable of donating a proton ( $\text{H}^+$ )
- A **Bronsted-Lowry base** is any species that is capable of accepting a proton such base requires a lone pair of electrons to bond to the  $\text{H}^+$ .
- Amphoteric (amphiprotic):** Substances which can act as acid as well as base are known as amphoteric.

### Conjugate Acid Base Pair

- The **conjugate base** of a Bronsted-Lowry acid is the species formed after an acid donates a proton. The **conjugate acid** of a Bronsted-Lowry base is the species formed after a base accepts a proton.

In a typical acid base reaction,  $\text{HX} + \text{B} \rightleftharpoons \text{X}^- + \text{HB}^+$

| Acid                                               |   | Base             |                      | Conjugate Acid                |   | Conjugate Base                                         |
|----------------------------------------------------|---|------------------|----------------------|-------------------------------|---|--------------------------------------------------------|
| HCl                                                | + | H <sub>2</sub> O | $\rightleftharpoons$ | H <sub>3</sub> O <sup>+</sup> | + | Cl <sup>-</sup>                                        |
| HSO <sub>4</sub> <sup>-</sup>                      | + | NH <sub>3</sub>  | $\rightleftharpoons$ | NH <sub>4</sub> <sup>+</sup>  | + | SO <sub>4</sub> <sup>-2</sup>                          |
| [Fe(H <sub>2</sub> O) <sub>6</sub> ] <sup>3+</sup> | + | H <sub>2</sub> O | $\rightleftharpoons$ | H <sub>3</sub> O <sup>+</sup> | + | [Fe(H <sub>2</sub> O) <sub>5</sub> (OH)] <sup>2+</sup> |

### Lewis Concept (electronic concept)

An **acid** is a molecule/ion that can accept an electron pair with the formation of a coordinate bond.

Such chemical species can act as Lewis Acid.

**Electron deficient molecules:** e.g., BF<sub>3</sub>, AlCl<sub>3</sub>, FeCl<sub>3</sub>

**Cations:** H<sup>+</sup>, Fe<sup>2+</sup>, Na<sup>+</sup>

A **base** is any molecule/ion that has a lone pair of electrons which can be donated. Such chemical species can act as Lewis base.

Base → (One electron pair donate)

**Molecules with lone pairs:** e.g., NH<sub>3</sub>, PH<sub>3</sub>, H<sub>2</sub>O, CH<sub>3</sub>OH

□ All negative charged ions e.g., CN<sup>-</sup>, Cl<sup>-</sup>.

### pH Values of a few Common Solutions

| Solution      | pH range  | Solution              | pH Range    |
|---------------|-----------|-----------------------|-------------|
| Coffee        | 4.5 – 5.5 | Lime water            | 12.0        |
| Milk          | 6.5       | Sodium hydroxide      | 14.0        |
| Tears         | 7.4       | Saliva                | 6.5 – 7.5   |
| Gastric juice | 1.0 – 3.0 | Pure water            | 7.0         |
| Vinegar       | 3.0       | Baking soda solution  | 8.4         |
| Acid rain     | 5.6       | House hold ammonia    | 11.9        |
| Sea water     | 9.0       | Bear                  | 4.0 – 5.0   |
| Lemon juice   | 2.2 – 2.4 | Blood                 | 7.36 – 7.42 |
| Tomato juice  | 4.1       | Washing soda solution | 10.5        |

### Ionic Product of Water (K<sub>w</sub>)

□ The product of the concentration of hydrogen ions or hydronium ions and hydroxide ions at any temperature in pure water or in any aqueous solution is known as the ionic product of water. It is represented by K<sub>w</sub>.

$$K_w = [H^+][OH^-]$$

$$K_w = [H_3O^+][OH^-]$$

$$pK_w = -\log K_w$$

□ For any aqueous solution, the sum of pH and pOH depends on the ionic product of water.

□ pH + pOH = pK<sub>w</sub> at any temperature

□ At 25° C, for any aqueous solution. pH + pOH = 14

□ The pH of pure water decreases as the temperature increases, similarly, the pOH also decrease.

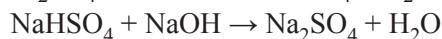
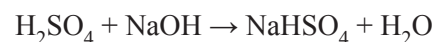
### Classification of Salts Based on their mode of Formation.

1. **Normal salt:** A salt which does not contain any replaceable H atoms or hydroxyl groups

**For Examples:** CaSO<sub>4</sub>, K<sub>3</sub>PO<sub>4</sub>, Na<sub>2</sub>SO<sub>4</sub>

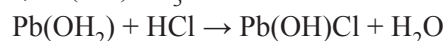
2. **Acidic salt:** A salt that contains replaceable hydrogen atoms is called an acidic salt.

**For Examples:** NaHSO<sub>4</sub>, KHCO<sub>3</sub> and KH<sub>2</sub>PO<sub>4</sub> are examples of acidic salts.



- 3. Basic salt:** A salt that contains replaceable hydroxyl group(s) is called a basic salt.

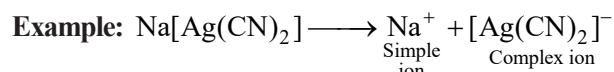
**For Examples:**  $\text{Cu}(\text{OH})\text{Cl}$ ,  $\text{Pb}(\text{OH})\text{Cl}$ ,  $\text{Pb}(\text{OH})\text{NO}_3$  etc



- 4. Double salt:** The salts which are obtained by the crystallisation of two simple salts from a mixture of their saturated solutions are known as double salts.

**For example:** Potash alum,  $(\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O})$ , Mohr's salt  $(\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O})$  and Dolomite  $(\text{CaCO}_3 \cdot \text{MgCO}_3)$  are double salts.

- 5. Complex Salt:** Such salts give one simple ion and a complex ion on dissociation.



## Classification of Salts Based on pH Values

- 1. Neutral salt solutions:** Salt solutions of strong acids and strong bases are neutral in nature and they have pH value equal to 7. They do not change the colour of litmus solution.

For e.g.,  $\text{NaCl}$ ,  $\text{NaNO}_3$ ,  $\text{Na}_2\text{SO}_4$  etc.

- 2. Acidic salt solutions:** Salt solutions of strong acids and weak bases are of acidic nature and they have pH value less than 7. They change the colour of blue litmus solution to red.

For e.g.,  $(\text{NH}_4)_2\text{SO}_4$ ,  $\text{NH}_4\text{Cl}$  etc.

In both these salts, the base  $\text{NH}_4\text{OH}$  is weak while the acids  $\text{H}_2\text{SO}_4$  and  $\text{HCl}$  are strong.

- 3. Basic salt solutions:** Salt solutions of strong bases and weak acids are of basic nature and they have pH value more than 7. They change the colour of red litmus solution to blue.

For e.g.,  $\text{K}_3\text{PO}_4$ ,  $\text{Na}_2\text{CO}_3$  etc.

In both these salts, the bases  $\text{KOH}$  and  $\text{NaOH}$  are strong while the acids  $\text{H}_3\text{PO}_4$  and  $\text{H}_2\text{CO}_3$  are weak.



## Multiple Choice Questions

- If the  $[\text{OH}^-] = 3 \times 10^{-14} \text{ M}$ , then the resulting solution is  
(a) Acidic (b) Basic  
(c) Neutral (d) Both (a) and (b)
- An unknown solution in a test tube was given to a student, when universal indicator solution is added to test tube, it turned out violet. The unknown solution is:  
(a) Vinegar solution (b) Iodine solution  
(c) Caustic Soda solution (d) Baking soda solution
- Name the gas that is evolved on reaction of dilute hydrochloric acid with sodium sulphite?  
(a) Sulphur trioxide (b) Hydrogen Sulphide  
(c) Sulphur dioxide (d) Carbon dioxide
- $\text{A} + \text{H}_2\text{O} \rightarrow \text{B}$ , B is used to remove grease stains from woolen clothes, Identify A and B respectively are  
(a)  $\text{NH}_3$ ,  $\text{NH}_4\text{Cl}$  (b)  $\text{NH}_3$ ,  $\text{NH}_4\text{OH}$   
(c)  $\text{NH}_4\text{OH}$ ,  $\text{NH}_3$  (d)  $\text{Cl}_2$ ,  $\text{HCl}$
- In cold countries, which of the following compound is usually spread on icy roads for melting ice?  
(a) Sodium bicarbonate  
(b) Rock salt  
(c) Sodium carbonate  
(d) Magnesium hydroxide
- The pH of three solutions X, Y and Z is 6, 4 and 8 respectively which of the correct option from the following?  
(a)  $\text{X} > \text{Y} > \text{Z}$  decreasing acidic strength  
(b)  $\text{Z} > \text{Y} > \text{X}$  increasing acidic strength  
(c)  $\text{Y} > \text{X} > \text{Z}$  decreasing acidic strength  
(d)  $\text{Z} > \text{Y} > \text{X}$  decreasing acidic strength
- Choose the correct oxide from following oxides which when dissolved in water gives a solution that will turn blue litmus to red?  
(a)  $\text{MgO}$  (b)  $\text{CaO}$   
(c)  $\text{CO}_2$  (d)  $\text{Na}_2\text{O}$
- The acid that is used by goldsmiths for cleaning gold and silver ornaments is:  
(a)  $\text{CH}_3\text{COOH}$  (b)  $\text{H}_3\text{PO}_4$   
(c)  $\text{HNO}_3$  (d)  $\text{H}_2\text{SO}_4$
- Which is not correct pair of conjugate acid and base from the following?  
(a)  $\text{HNO}_3$  and  $\text{NO}_3^-$  (b)  $\text{NH}_4^+$  and  $\text{NH}_3$   
(c)  $\text{H}_3\text{PO}_2$  and  $\text{H}_2\text{PO}_2^-$  (d)  $\text{H}_2\text{SO}_4$  and  $\text{SO}_4^{2-}$
- A man is using excess quantities of chemical fertilizers in fields for a number of years. The crop productivity gradually began to decrease in spite of using chemical fertilizers.  
Which of the following are correct statements?  
(i) The soil has become more acidic in nature  
(ii) The soil has become more alkaline in nature  
(iii) The productivity of crop is enhanced by adding slaked lime to the soil.  
(iv) The productivity of crop is enhanced by adding some organic matter.  
(v) The productivity of crop can be enhanced by adding  $\text{NaCl}$ .  
(a) (i) and (v) (b) (i) and (iii)  
(c) (ii) and (v) (d) (ii) and (iv)
- pH of a 0.001 M  $\text{NaOH}$  solution is  
(a) 2 (b) 10  
(c) 3 (d) 11
- When  $\text{HCl(g)}$  is passed through water  
(a) It gives both  $\text{H}^+$  ions and  $\text{OH}^-$  in the solution  
(b) It ionises in solution  
(c) It does not ionise in solution  
(d) None of the above
- When  $\text{HCl}$  gas is prepared on a humid day, the gas is generally passed through the guard tube containing  $\text{CaCl}_2$ . The role of  $\text{CaCl}_2$  taken in the guard tube is to  
(a) Absorb  $\text{Cl}^-$  ions from the evolved gas  
(b) Absorb the gas  
(c) Absorb moisture from the gas  
(d) Absorb the evolved gas
- When copper oxide and dilute hydrochloric acid react, the colour changes to which of the following colour?  
(a) Black (b) Bluish-green  
(c) White (d) Yellow
- Arrhenius acid gives  
(a)  $\text{H}^+$  in water (b)  $\text{OH}^-$  in water  
(c) Both of the above (d) None of these

16. Which of the following is considered as a strong base?
- Ammonium hydroxide ( $\text{NH}_4\text{OH}$ )
  - Sodium hydroxide ( $\text{NaOH}$ )
  - Water ( $\text{H}_2\text{O}$ )
  - Nitric acid ( $\text{HNO}_3$ )
17. Which of the following is considered as a strong acid?
- Acetic acid
  - Maleic acid
  - Nitric acid
  - Tartaric acid
18. Which of the following aqueous solution of salts will turns the red litmus to blue?
- $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
  - $\text{NaHCO}_3$
  - $\text{Na}_2\text{CO}_3$
  - All
19. Soda-acid fire extinguishers extinguishes prevent fire
- By cutting the supply of air.
  - By removing the combustible substance
  - By increasing the ignition temperature
  - All the above
20. When pH paper is put in distilled water its colour changes to green. Then some common salt is added to water and pH paper is tested in this solution. Now the change colour of pH paper in this case is:
- Green
  - Pink
  - Red
  - Blue
21. A is an aqueous solution of acid and B is an aqueous solution of base. Water is added to both separately. Then what are the changes noticed in pH?
- pH of A increases while that of B decreases till neutral
  - pH of B decreases while that of A increases till neutral
  - pH of A and B increases
  - pH of A and B decreases
22. Which of the following is not a example of hydrated salt ?
- Epsom salt
  - Baking soda
  - Washing soda
  - Blue vitriol
23. Which of the following salts does not contain water of crystallisation?
- Blue vitriol
  - Baking soda
  - Washing soda
  - Gypsum
24. Equal volumes of hydrochloric acid and sodium hydroxide solutions of same concentration are mixed and the pH of the resulting solution is checked with a pH paper. What would be the colour obtained? (You may use colour guide given in Figure.)
- |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
- Red
  - Yellow
  - Yellowish green
  - Blue
25. Which of the following is not a mineral acid?
- Hydrochloric acid
  - Citric acid
  - Sulphuric acid
  - Nitric acid
26. The acid which contains four hydrogen atoms \_\_\_\_\_.
- Formic acid
  - Sulphuric acid
  - Nitric acid
  - Acetic-acid



## Competitive Corner

1. Which salts are responsible for yellow colour of Taj Mahal in Agra due to Acid rain (2022)
- $\text{CaCl}_2$  &  $\text{CaSO}_4$
  - $\text{Ca}(\text{NO}_3)_2$  &  $\text{CaSO}_4$
  - $\text{Ca}(\text{NO}_3)_2$  &  $\text{BaSO}_4$
  - $\text{CaSO}_4$  &  $\text{BaCl}_2$
2. One of the following is not an organic acid. This is (Jharkhand 2018)
- Ethanoic acid
  - Formic acid
  - Citric acid
  - Carbonic acid
3. The indicators which turn red in acid solution are: (Jharkhand 2018)
- Turmeric and Litmus
  - Phenolphthalein and Methyl Orange
  - Litmus and Methyl Orange
  - Phenolphthalein and Litmus
4. Which of these can be used as olfactory indicator? (Delhi 2018)
- Vanilla
  - Onion
  - Clove
  - All the above three
5. What happen, when methyl orange solution mixed with HCl.

- (a) Solution becomes yellow  
(b) Solution becomes red  
(c) Solution becomes blue  
(d) Solution becomes pink
6. Which of these salts will give acidic solution?  
(Delhi 2018)
- (a)  $\text{Na}_2\text{CO}_3$  (b)  $\text{NaCl}$   
(c)  $\text{NH}_4\text{Cl}$  (d)  $\text{HCOONa}$
7. pH is define as (Delhi 2018)
- (a)  $-\log [\text{H}_3\text{O}^+]$  (b)  $-\log [\text{H}_2\text{O}]$   
(c)  $-\log [\text{OH}^-]$  (d)  $-\log [\text{H}^+][\text{OH}^-]$
8. Baking powder is a mixture of: (Haryana 2017)
- (a) Sodium carbonate & sodium hydrogen carbonate  
(b) Sodium carbonate & acetic acid  
(c) Sodium hydrogen carbonate & methanoic acid  
(d) Sodium hydrogen carbonate & tartaric acid
9. Which one of the following oxide is insoluble in water?  
(Delhi 2020)
- (a)  $\text{Na}_2\text{O}$  (b)  $\text{CuO}$   
(c)  $\text{K}_2\text{O}$  (d)  $\text{CaO}$
10. Which of the following oxide turns red litmus into blue?  
(Delhi 2020)
- (a)  $\text{SO}_2$  (b)  $\text{CO}_2$   
(c)  $\text{NO}_2$  (d)  $\text{K}_2\text{O}$
11. The value of pOH of rain water is: (Tamil Nadu 2020)
- (a) 7 (b) 6  
(c) 8.5 (d) 9.5
12. Name the element which is common to all acids?  
(Madhya Pradesh 2020)
- (a) Sulphur (b) Chlorine  
(c) Nitrogen (d) Hydrogen
13. Dilution is the process of (Madhya Pradesh 2020)
- (a) Mixing acid with water  
(b) Mixing strong acid with strong base  
(c) Mixing acid or base with water  
(d) Mixing strong acid with weak base
14. What type of reaction takes place when an acid is dissolved in water? (Madhya Pradesh 2020)
- (a) Exothermic (b) Endothermic  
(c) Substitution (d) Elimination
15. Select the organic acid from the following:  
(Jharkhand 2020)
- (a) Hydrochloric acid  
(b) Nitric acid  
(c) Sulphuric acid  
(d) Citric acid

16. Match List-I with List-II and select the correct answer by using the codes given below the list:

(Chandigarh 2002)

| List-I<br>(Named of Acid) |               | List-II<br>(Source) |          |
|---------------------------|---------------|---------------------|----------|
| P.                        | Lactic acid   | (i)                 | Tamarind |
| Q.                        | Oxalic Acid   | (ii)                | Curd     |
| R.                        | Acetic Acid   | (iii)               | Tomato   |
| S.                        | Tartaric Acid | (iv)                | Vinger   |

- (a) P-(i), Q-(iv), R-(ii), S-(iii)  
(b) P-(ii), Q-(i), R-(iv), S-(iii)  
(c) P-(ii), Q-(i), R-(iii), S-(iv)  
(d) P-(ii), Q-(iii), R-(iv), S-(i)
17. Indicators are (Andhra Pradesh 2020)
- (i) Weak acids or weak bases  
(ii) Strong acids or strong bases  
(iii) They help to find acidic and basic nature of liquids  
(iv) Most of them are organic compounds
- (a) (i), (ii), (iv) (b) (i), (iii), (iv)  
(c) (ii), (iii), (iv) (d) (i), (ii), (iii)
18. For testing pH of the soil, a student mixed a sample of soil thoroughly with distilled water and allowed the container to sit undisturbed. Blue litmus turned red on dipping in the soil-water mixture. Which of the following would change the colour of the red litmus to blue?  
(Odisha 2020)
- (a) Common salt (b) Vinegar  
(c) Baking powder (d) Hydrochloric acid
19. Which acid is present in tomato? (Gujarat 2019)
- (a) Citric Acid (b) Acetic Acid  
(c) Oxalic Acid (d) Tartaric Acid
20. Bleaching powder is represented by formula  
(Telangana 2019)
- (a)  $\text{NaHCO}_3$  (b)  $\text{Na}_2\text{CO}_3$   
(c)  $\text{CaOCl}_2$  (d) None
21. The pH of acid rain is (Delhi 2019)
- (a) less than 5.6 (b) more than 5.6  
(c) equal to 5.6 (d) more than 6.6
22. A student added a drop of universal indicator to 1.00mL of given solution and found that a green colour is produced, The pH value of the solution will be
- (a) 7 – 9 (b) 0 – 3  
(c) 10 – 12 (d) 4 – 6

23. The chemical formula of Gypsum is: (MP 2017)

- (a)  $\text{CaCO}_3 \cdot 5\text{H}_2\text{O}$  (b)  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$   
 (c)  $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$  (d)  $\text{CaCO}_3 \cdot \frac{1}{2} \text{H}_2\text{O}$

24. The correct order of increasing pH values of the aqueous solutions of baking soda, rock salt, washing soda and slaked lime is (NTSE 2018)

- (a) Baking soda < Rock salt < Washing soda < Slaked lime  
 (b) Rock salt < Baking soda < Washing soda < Slaked lime  
 (c) Slaked lime < Washing soda < Rock salt < Baking soda  
 (d) Washing soda < Baking soda < Rock salt < Slaked lime

25. Equal volumes of solutions containing 1 mole of an acid and 1 mole of a base respectively are mixed. Which of these mixture will give pH more than 7? (NTSE 2020)

- (a) Sodium hydroxide + Acetic acid  
 (b) Potassium hydroxide + Sulphuric acid  
 (c) Ammonium hydroxide + Sulphuric acid  
 (d) Sodium hydroxide + Hydrochloric acid

26. Sulphur powder is heated on a spatula. A piece of both, moist blue and red litmus papers are brought one by one near the gas evolved during heating. The action of gas on the moist litmus papers will be: (NTSE 2021)

- (a) No change in colour in both the litmus papers.  
 (b) Blue litmus paper becomes red.  
 (c) Red litmus paper becomes blue.  
 (d) Blue litmus paper turns black.

27. You are provided with aqueous solutions of three salts-A, B and C, 2-3 drops of blue litmus solution, red litmus solution and phenolphthalein were added to each of these solution in separate experiments. The change in colours of different indicators were recorded in the following table. (NTSE 2019)

| Sample | With blue litmus solution | With red litmus solution | With phenolphthalein solution |
|--------|---------------------------|--------------------------|-------------------------------|
| A      | No change                 | No change                | No change                     |
| B      | Turns red                 | No change                | No change                     |
| C      | No change                 | Turns blue               | Turns pink                    |

On the basis of above observations, identify A, B and C from the following options:

- (a) A =  $\text{NH}_4\text{Cl}$ , B =  $\text{NaCl}$ , C =  $\text{CH}_3\text{COONa}$   
 (b) A =  $\text{NH}_4\text{Cl}$ , B =  $\text{CH}_3\text{COONa}$ , C =  $\text{NaCl}$   
 (c) A =  $\text{NaCl}$ , B =  $\text{NH}_4\text{Cl}$ , C =  $\text{CH}_3\text{COONa}$   
 (d) A =  $\text{CH}_3\text{COONa}$ , B =  $\text{NH}_4\text{Cl}$ , C =  $\text{NaCl}$

28. When four dilute solutions of (IOQJS 2019)

- (i) vinegar  
 (ii) common salt,  
 (iii) caustic soda and  
 (iv) baking soda are tested with universal indicator which will be the correct observation  
 (a) (i)-Green, (ii)-Violet, (iii)-Blue, (iv)-Red  
 (b) (i)-Green, (ii)-Blue, (iii)-Violet, (iv)-Red  
 (c) (i)-Orange, (ii)-Green, (iii)-Violet, (iv)-Blue  
 (d) (i)-Orange, (ii)-Violet, (iii)-Green, (iv)-Blue

29. Shaila took about 10 cm<sup>2</sup> of a diluted Potassium hydrogen carbonate solution in a test tube. To this solution she added few drops of universal indicator. The colour of the solution turned: (JS<sub>511</sub>)

- (a) orange  
 (b) green  
 (c) blue  
 (d) yellow

30. Which of the following represents the correct order of the acidic strength for equimolar aqueous solutions of HCl,  $\text{H}_2\text{SO}_4$ ,  $\text{NH}_4\text{OH}$  and NaOH [NTSE 2014]

- (a)  $\text{HCl} < \text{NH}_4\text{OH} < \text{NaOH} < \text{H}_2\text{SO}_4$   
 (b)  $\text{NH}_4\text{OH} < \text{NaOH} < \text{H}_2\text{SO}_4 < \text{HCl}$   
 (c)  $\text{HCl} < \text{H}_2\text{SO}_4 < \text{NH}_4\text{OH} < \text{NaOH}$   
 (d)  $\text{NaOH} < \text{NH}_4\text{OH} < \text{HCl} < \text{H}_2\text{SO}_4$

31. An element X reacts with dilute  $\text{H}_2\text{SO}_4$  as well as with NaOH to produce salt and  $\text{H}_2(\text{g})$ . Hence, it may be concluded that: [NTSE 2016]

- (i) X is an electro-positive  
 (ii) Oxide of X is basic in nature.  
 (iii) Oxide of X is acidic in nature.  
 (iv) X is an electronegative element.  
 (a) (i), (ii), (iii)  
 (b) (iv), (i), (ii)  
 (c) (ii), (iii), (iv)  
 (d) (iii), (iv), (i)



## Explanations



### School Level

#### Quick Recall

##### Fill in the Blanks

1. blue, red    2. more/greater    3. 7
4.  $H^+$     5. Baking soda
6. Water of crystallisation    7. acid (HCl)
8.  $CaSO_4 \cdot Cl_2 \cdot H_2O$     9.  $Na_2CO_3 \cdot 10H_2O$
10. Basic

##### True and False Statements

1. False    2. True    3. True    4. True
5. False    6. False    7. False    8. True
9. False    10. True

##### Match the Followings

1. (c)    2. (b)    3. (a)    4. (a)    5. (c)

##### Assertion & Reason Type Questions

1. (a) Sodium carbonate reacts with excess hydrochloric acid to form sodium chloride, water and carbon dioxide. In this reaction, bubbles of carbon dioxide are observed.
2. (b) Ammonia ( $NH_3$ ) is considered as base but it does not contain hydroxide  $OH^-$  ions. It is a base because it can donate a lone pair of electron (according to Lewis concept). But in aqueous solution, it will act as base, accept  $H^+$  from water to form  $NH_4^+$  and  $OH^-$ .
3. (c) Ammonia gas, which is alkaline, turns the red litmus paper blue.
4. (c) The body part where an ant bite is treated with moist baking soda.
5. (d) Concentrated  $H_2SO_4$  can be diluted by adding acid dropwise to water. This will minimise effect of the heat released during the reaction.  
 $H_2SO_4$  has strong affinity for water because acid from hydrates with water.

##### Statement Type Questions

1. (a) Both statements I and II are correct.
2. (a) Both statements I and II are correct.

3. (d) All metal oxides are not basic in nature.  $Al_2O_3$  is a metallic oxide and is amphoteric in nature. Lemon juice is more acidic than tomato juice.
4. (c) Aqueous solution of glucose and alcohol do not give  $H^+$  ions.
5. (b) Both statements are incorrect. Tooth decay starts when the pH of the mouth is lower than 5.5. Tooth enamel is made up of calcium hydroxyapatite (a crystalline form of calcium phosphate) and is the hardest substance in the body.

#### Multiple Choice Questions

1. (a)    2. (c)    3. (b)    4. (d)    5. (b)
6. (d)    7. (b)    8. (d)    9. (b)    10. (d)
11. (a)    12. (a)    13. (b)    14. (a)    15. (d)

#### Subjective Questions

##### Very Short Answer Type Questions

1. The colour of phenolphthalein in milk of magnesia is pink.
2. Ammonia dissolves in water to form ammonium hydroxide ( $NH_4OH$ ) which is a base and it turns red litmus into blue.  
$$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$$
Dry  $NH_3$  gas will not change into  $OH^-$  ions and cannot turn red litmus to blue.
3. The taste of the vegetable soup is likely to be slightly sour because it is weakly acidic.
4. Acidic nature will increase with increase in concentration of  $H_3O^+$  or  $H^+$  ions.
5. Hydrangea, Petunia are the flowers whose colour of petal can be used as indicator.
6. Formic acid is present in honey bee sting and hair of nettle leaves.
7. If a solution has higher concentration of  $H^+$ , then lower will be pH value of the solution.
8. Bitter melon extract changes red litmus blue. It is basic in nature.
9. Venus.
10. Beetroot extract is an acid base indicator used at home.
11. Carbon dioxide gas is released when  $NaHCO_3$  reacts with dilute HCl.

12. The salt is sodium hydrogen carbonate ( $\text{NaHCO}_3$ ).  
The chemical equation used for its formation is  

$$\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$$
 Its two uses are:  
 (a) For making baking powder.  
 (b) Used as an antacid.

### Short Answer Type Questions

- (a) Oxalic acid is present in tomato  
(b) Tartaric acid is present in tamarind
- (a) Lemon and orange  
(b) Tomatoes and Guava.  
(c) Sour milk (curd)  
(d) Tamarind.
- (a) The acid is to be added slowly in water to avoid the mixture to be splashed out. The reaction is highly exothermic in nature therefore, constant stirring should be done.  
(b) The process is known as dilution.
- Solution 'X' is acidic in nature as it turns orange when a drop of it falls on a pH paper. Its pH value will be in the range of 3 to 5 and is moderately acidic.  
Solution 'Y' is basic in nature as it turns into blue colour when a drop of it falls on a pH paper. Its pH value ranges from 8 to 11 and is moderately basic.

| Given substance   | Given pH |           |
|-------------------|----------|-----------|
| Acetic acid       | 3        | Correct   |
| Water             | 12       | Incorrect |
| Lemon juice       | 4        | Correct   |
| $\text{NaHCO}_3$  | 8        | Correct   |
| $\text{NaOH(aq)}$ | 14       | Correct   |

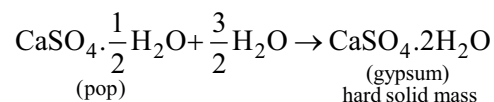
Correct pH for water will be 7, because it is neutral in nature.

6. Pale green is the colour of  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$  crystals. Yes, colour will change. It becomes dirty white on heating.
- $$\underset{\text{Pale green}}{\text{FeSO}_4 \cdot 7\text{H}_2\text{O}} \xrightarrow{\text{heat}} \underset{\text{Dirty white}}{\text{FeSO}_4} + 7\text{H}_2\text{O}$$
7. Acidity is caused by excess of HCl in stomach. Antacid which is basic in nature. It neutralizes the effect of excess of acid in stomach and gives relief from pain caused by acidity.

8. (a) Chemical name is calcium sulphate hemihydrate and chemical formula is  $(\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O})$

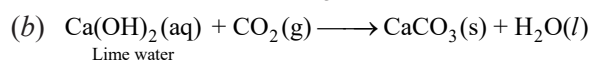
(b) The powder is prepared by heating gypsum (Calcium sulphate dihydrate) at  $100^\circ\text{C}$ .

(c) The Balanced chemical equation is given as follows-

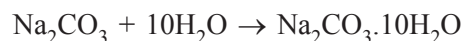


(d) Plaster of Paris is used in preparing mould, statues and artificial decorative items.

9. (a) Lime water will turn milky due to formation of calcium carbonate ( $\text{CaCO}_3$ ).



10. By recrystallisation of sodium carbonate, washing soda can be obtained.



It is a basic salt.

The permanent hardness of water can be removed by it.

11. Sodium hydrogencarbonate is a salt of sodium hydroxide NaOH (strong base) and carbonic acid  $\text{H}_2\text{CO}_3$  (weak acid).

Thus it is basic salt. Common name is baking soda.

12. Preparation



**Uses:** It is used as an antacid in medicine to remove acidity.

13. Water of crystallisation is the fixed number of water molecules present in one formula unit of a salt.

For example,  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ,  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

Take one gram of powder copper sulphate crystals ( $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ) which are blue in colour in a dry China dish. It loses water molecules and a white powder ( $\text{CuSO}_4$ ) is obtained. Cool and add a few drops of water to the white powder. Blue colour is restored because of the formation of copper sulphate hydrated ( $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ).

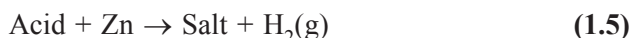
We conclude that water of crystallisation makes a difference in the state and colour of the compound.

### Long Answer Type Questions

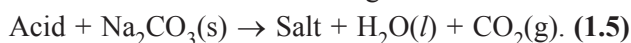
1. Since liquid 'X' changes its odour when a cloth strip is dipped in onion juice, this indicates the acidic nature of the substance, and the indicator it acts as will be an olfactory indicator. Also, since the liquid

'X' turns blue litmus to red, the liquid is an acid. (2)

(a) When this acid reacts with the metal Zinc, it produces salt with the evolution of hydrogen gas.



(b) When this acid reacts with solid sodium carbonate, the products formed are: Salt, water and the release of carbon dioxide gas.

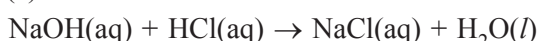


2. (a) Indicator is a substance which gives different colour or odour in acid and basic medium e.g., litmus and turmeric are indicators obtained from plants. (2)



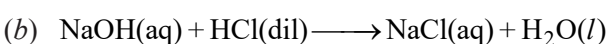
Solution will turn phenolphthalein pink and red litmus paper blue. (2)

(c) Sodium chloride and water are formed: (1)

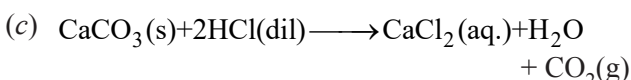


3. (a)  $\text{Mg}(\text{s}) + 2\text{HCl}(\text{dil}) \longrightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$

Hydrogen gas will be released in this reaction (1.5)



Sodium chloride and water will be formed (1.5)



Crushed egg shell contain  $\text{CaCO}_3$  which reacts with dil HCl to produce brisk effervescence due to  $\text{CO}_2$  and eggshells become soft. (2)

4. (a) Tap water contains ions due to which electricity is conducted, distilled water does not contain any ions to conduct electricity. (1)

(b) Dry HCl gas will not form ions but dil HCl gives  $\text{H}^+$  and  $\text{Cl}^-$ . (1)

(c) Baking soda solution will not allow milk to change into lactic acid that makes milk taste sour. (1)

(d) Adding water to acid is highly exothermic in nature, that can cause burn if splashed out. Thus, water is added very slowly to acid. (1)

(e) Ammonia dissolves in water to give  $\text{NH}_4\text{OH}$  and forms  $\text{OH}^-$ . Thus, it is basic in nature. (1)

5. (a) Common salt is obtained from the combination of the acid hydrochloric acid (HCl) and the base sodium hydroxide (NaOH).

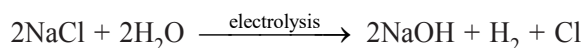
The formula of the salt is NaCl.

**Chemical Name:** Sodium Chloride

It is obtained from seawater. (2)

(b) Rock salt is the common name for the mineral "halite". Its chemical formula is NaCl. It may be white or light blue, or yellow, depending upon the impurities present in it. (1.5)

(c) When electricity is passed through brine (an aqueous solution of sodium chloride), it decomposes to form sodium hydroxide. (1.5)



6. (a) Baking soda is sodium hydrogencarbonate.

On heating, it is converted into sodium carbonate which is bitter in taste

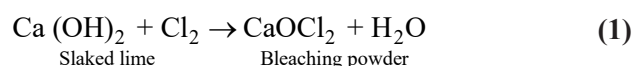


(b) Baking soda can be converted into baking powder by the addition of appropriate amount of tartaric acid to it. (1.5)

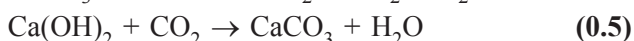
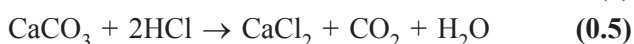
(c) The role of tartaric acid is to neutralise sodium carbonate so that the cake will not taste bitter. (1.5)

7. The gas evolved at anode during electrolysis of brine is chlorine gas(G). (1)

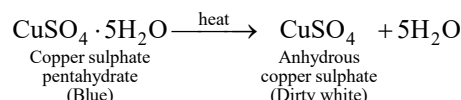
When chlorine gas is passed through dry  $\text{Ca}(\text{OH})_2$  (Y) produces bleaching powder (Z) used for disinfecting drinking water. (1)



Since Y and Z are calcium salts, therefore X is also a calcium salt and is calcium carbonate. (1)



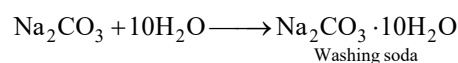
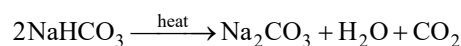
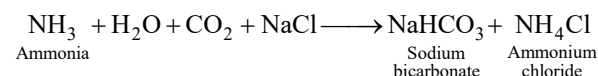
8. (a)  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  is a blue crystalline solid which becomes dirty white on heating because of loss of water molecules and it will become amorphous  $\text{CuSO}_4$ .



$\text{CuSO}_4$  acquires its colour by absorbing water from atmosphere and becomes blue again in colour.



(b)  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$  is compound having 10 $\text{H}_2\text{O}$  molecule. It is called sodium carbonate decahydrate or washing soda. It is prepared by passing  $\text{CO}_2$  gas through saturated solution of ammonical brine.





10. (d)  $K_2O$  is a metal oxide, thus will turn red litmus into blue.
11. (c) We know that  
 $pK_w = pH + pOH$   
 $pK_w = 14$   
 $pH \text{ of rainwater} = 5.6$   
Hence,  $pH = 14 - 5.6 = 8.4$  or  $8.5$  (approx)
12. (d) Hydrogen is an element common to all acids. For example,  $H_2SO_4$  (Sulphuric acid),  $HNO_3$  (Nitric Acid), and  $CH_3COOH$  (Acetic acid) all have hydrogen atoms in it.
13. (c) Acids and bases are mostly soluble in water, and they can be diluted by adding the required amount of water and this process of adding water to an acid or base is known as dilution.
14. (a) Dilution is an exothermic reaction.
15. (d) Hydrochloric acid, Nitric acid, and Sulphuric acid are all mineral acids. But Citric acid is an example of organic acid.
16. (d) P-(ii), Q-(iii), R-(iv), S-(i).
17. (b) Indicators are weak acids or weak bases that help to find acidic and basic nature of liquids, and most of them are organic compounds.
18. (c) Baking powder ( $NaHCO_3$ ) is basic in nature, and it will turn red litmus to blue.
19. (c) Oxalic Acid is present in tomato.
20. (c)  $CaOCl_2$  is the chemical formula of bleaching powder.
21. (a) The pH of acid rain is less than 5.6.
22. (a)  $7 - 9$
23. (b) The chemical formula of gypsum is  $CaSO_4 \cdot 2H_2O$
24. (b) Rock salt =  $NaCl$ ,  
Baking soda =  $NaHCO_3$   
Washing soda =  $Na_2CO_3$   
Slaked lime =  $Ca(OH)_2$   
Increasing order of pH value is  
Rock Salt < Baking Soda < Washing Soda < Slaked lime.
25. (a) Strong Base + Weak Acid  $\rightarrow$  Basic salt.
26. (b) When the sulphur powder is heated on a spatula, the gas evolved is sulphur dioxide ( $SO_2$ ). Since  $SO_2$  is an oxide of non-metal so it will be acidic in nature and will turn blue litmus paper red.
27. (c) A =  $NaCl$  (Neutral in nature)  
B =  $NH_4Cl$  (Acidic in nature)  
C =  $CH_3COONa$  (Basic in nature)
28. (c) (i)-Orange, (ii)-Green, (iii)-Violet, (iv)-Blue
29. (c) The colour of the solution turned blue.
30. (d) The correct order of acidic strength is given as:  
 $NaOH < NH_4OH < HCl < H_2SO_4$   
 $H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$   
 $HCl \longrightarrow H^+ + Cl^-$   $H_2SO_4$  produces more no. of  $H^+$  ions on dissociation and thus stronger acid than  $HCl$ .  
 $NaOH \longrightarrow NaOH + OH^-$   
 $NH_4OH \rightleftharpoons NH_4^+ + OH^-$   
Both  $NH_4OH$  and  $NaOH$  will not give  $H^+$  ions on dissociation. But,  $NH_4OH$  partially dissociates and  $NaOH$  completely dissociate to give  $OH^-$  ions. Thus,  $NaOH$  is more stronger base and weaker acid than  $NH_4OH$ .
31. (a) Element X will react with both acid and base. It shows that element X has amphoteric nature and it is an electropositive element.